

Patents, Papers, Parts, and Planet

Exploration of the Innovation Space in Synthetic Biology

Zakhary Roth

Advisors

Claudia Doblinger, Technical University of Munich

Frank Neffke, Complexity Science Hub

Abstract

Science and technology has advanced to the point where we can treat DNA as a programming language and living organisms as hardware. Cells are factories of protein production, and we can enter into their insides and fine-tune the machinery that makes them run. With standardized genetic parts and widely used assembly protocols, life itself has become a series of tools to optimize production. This is a new age of biotechnological innovation, with tools and standards spread around the world, and a lower barrier of entry than ever before to genetic engineering. This project studies different forms of research in Synthetic Biology, student projects and academic publications, to see if geography plays a role in determining the focus of research.

Are students, academics, and private industry researching the same topics in the cities they live and work? Trends sweep across fields of study, but conventional wisdom holds that certain areas will collectively specialize in specific research areas or industries. The International Genetic Engineering Machine, or iGEM, is a nonprofit that advances Synthetic Biology (SynBio) education through the curation of an annual student research competition, and the data from this competition provides a potential wealth of insights into the process of innovation in this industry. We study the corpus of iGEM project data and compare it empirically to a body of academic SynBio publications, to see if geographically-influenced semantic relatedness can be established. Relatedness between iGEM projects, academic research, and patent abstracts is calculated, established, and empirically tested. Clusters are identified in the semantic space, and those related to carbon capture are inspected more in-depth. The research landscapes of various cities are compared, and policy suggestions are made.

Contents

List of Figures	iii
List of Tables	vi
List of Appendices	viii
List of Abbreviations	ix
List of Symbols	x
1 Declaration of Authorship	1
2 Statement on AI use	2
3 Introduction	3
4 Background	6
4.1 Innovation, research, and the role of the state in the Bioeconomy	6
4.2 A field worth watching	8
4.3 iGEM Projects	9
4.4 BioBricks and the standardized parts registry	9
4.5 Booze Bugs	10
4.6 Algae, oxygen, and carbon capture	11
4.7 What iGEM leaves behind	12
4.8 Why might local alignment exist?	14
4.9 The geography of innovation	15
4.10 Measuring relatedness	17
4.11 Semantic Space	20
4.12 Three artifacts, One Space	21
5 Data	22
5.1 A shared corpus	22
5.2 Patents	23
5.3 Papers	24
5.4 Projects	25
5.5 Parts	26
5.6 Linking the layers and fine-tuning	28
5.7 Embedding, projection, and clustering	30
5.8 The carbon-capture slice	32

6	Results	34
6.1	Project-Paper Basic Alignment	34
6.2	Better Alignment through Clusters	36
6.3	Lead-lag structure	48
6.4	Carbon-capture alignment	53
6.5	Discussion	56
6.6	Limitations	58
7	Conclusion	59
A	Custom Claude Code skills	64
B	OpenAlex keyword query	67

List of Figures

1	Conceptual model of the three-artifact innovation ladder. Student iGEM projects introduce early-stage biological designs at time t ; academic papers develop and disseminate those ideas roughly three years later. patents mark downstream commercial translation at a further lag. All three activities co-occur within a shared urban institutional environment whose role is the central object of study. Arrows label the proposed mechanism; the sequence is the hypothesis this thesis tests.	6
2	Geographic distribution of synthetic biology artifacts across cities. Each panel shows one artifact type: iGEM student projects (left), academic papers (centre), and patents (right). Circle size is proportional to the square root of artifact count in that city. Projects are the most geographically concentrated, reflecting the university-competition structure of iGEM; papers reach more cities because a broader range of institutions publish; patents are the sparsest, concentrated in established biotechnology hubs. Annotation in each panel reports the total number of geocoded cities and artifacts. The pattern confirms that the corpus spans multiple continents, though it is weighted toward North America and Europe by the structure of all three data sources.	23
3	Data pipeline and artifact counts at each processing stage. Three independent streams — iGEM Registry, OpenAlex, and the Oldham et al. synthetic biology patent corpus — are geocoded independently, then linked via cross-type citation pairs for fine-tuning. The fine-tuned SPECTER2 model embeds all 17,826 artifacts into a shared 768-dimensional semantic space, which is then projected and clustered to yield 84 topic clusters. City-level Jaccard overlap is computed for the 387 cities where at least two artifact types are co-present. Geocoding attrition is below 1.5% for each stream; the 21,959 project-to-paper wiki-DOI links provide the primary fine-tuning signal for cross-type alignment.	29
4	The corpus in two dimensions. Every project, paper, and patent is a point, placed by UMAP so that semantically similar artifacts fall near one another. Colour marks the artifact type.	31
5	The same map, with topic clusters labelled. Each labelled region is a cluster found by HDBSCAN and named from its most representative documents. The cyanobacterial and fermentation clusters that make up the carbon-capture case study sit close together.	33
6	Why the centroid measure fails. Project-to-paper centroid overlap climbs with the number of documents a city has, so the measure mostly tracks city size rather than local relatedness.	37

7	The central result. The observed rate at which a city’s projects and papers share topics (marked line) sits far above the distribution produced by reshuffling projects across cities (histogram), which holds city size and topic popularity fixed. The gap is the size-independent local relatedness.	38
8	Leave-one-country-out. Recomputing the excess with each country removed in turn leaves it essentially unchanged; no single national system drives the result.	40
9	All three pairwise alignments across the full k -sweep (left: permutation excess over null; right: p -values). Project×paper and paper×patent both stay significant at every reasonable resolution. Project×patent never crosses the significance threshold (dashed line, right panel), establishing the ladder structure in which academic papers mediate between student projects and patents.	41
10	Varying the document threshold. As the minimum documents per city is raised and lowered, the excess persists, and shows more cleanly in the deeper cities. . .	43
11	Tripartite difference-in-differences: estimated $\hat{\beta}$ coefficients with 95% city-clustered confidence intervals for all three ordered pairs. A positive coefficient means that within a city and topic, the upstream artifact type’s share in a five-year period predicts a higher downstream share in the same period, after city baselines and topic trends are differenced out. Paper → patent is positive and significant ($\hat{\beta} = +0.355$, $p = 0.028$, $N = 44$ cities). Project → paper is not significant in this smaller sample ($p = 0.19$, $N = 46$ cities), consistent with insufficient power relative to the 161-city bipartite analysis. Project → patent uses only ten cities and is not interpretable.	49
12	Lead-lag profiles for the two adjacent pairs in the innovation ladder, with 95% bootstrap confidence intervals. Top: project activity at year t vs. paper activity at year $t + k$; positive k means papers come after projects. Bottom: paper activity at year t vs. patent activity at year $t + k$. Profiles are relatively flat across lags; the permutation tests (Figure 13) do not find a statistically significant directional asymmetry at this sample size.	50
13	Permutation tests of lead-lag asymmetry (500 within-city year shuffles), for both pairs. Each row is one pair; each column is one test statistic. Left column: the slope of the lead-lag profile against the null distribution. Right column: the contrast $\text{mean}(\ell > 0) - \text{mean}(\ell < 0)$ against its null. In both pairs the observed statistics fall inside the null distribution, indicating that the tripartite corpus does not produce a statistically significant temporal asymmetry at this sample size. .	51
14	Two independent views agree. City-pair similarity measured from the mix of BioBrick part classes (never seen by the language model) correlates with city-pair similarity measured from semantic topic profiles. The relatedness is anchored outside the embedding space.	54

15	The carbon-capture network. Projects, papers, and patents in the case-study clusters, linked by the citations and BioBrick connections recovered during corpus construction. The cyanobacterial work forms a distinct, tightly connected core.	55
16	City centrality in carbon capture. Cities ranked by how connected their carbon-capture work is within the network, rather than by volume alone. Cyanobacteria-focused cities lead.	56

List of Tables

- 3 Robustness check: excess co-membership under base vs. fine-tuned SPECTER2 across six neighbourhood sizes k . Each cell reports excess co-membership (observed minus null-model mean) from a permutation test ($n = 2,000$ shuffles). n = number of cities with at least one artifact of each type. 42
- 4 The first city-level alignment regression, and why it retired the centroid measure. Columns (1)–(2) regress the centroid overlap (`semantic_overlap`, the cosine between a city’s paper- and project-embedding centroids) on city size; column (3) regresses the cluster co-membership overlap (`cluster_overlap`, the cosine between the two cluster-frequency vectors) on the same size controls. Size explains most of the centroid measure and almost none of the cluster measure. Size is total document counts in (1)–(2) and clustered (non-noise) counts in (3), which is why the cluster sample is the 60 cities with ≥ 8 clustered documents of each type. Coefficients are not comparable across measures (the two overlaps live on different scales); the readable contrast is the R^2 and the size significance. Standard errors in parentheses: HC3 in (1) and (3), clustered by country in (2). $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ 44
- 5 Size-confound robustness. Dependent variable: city-level cluster overlap (cosine between paper and project cluster-frequency vectors; sample restricted to cities with ≥ 8 non-noise documents of each type, following Section 8.1 of the analysis). Specifications add controls sequentially. HC3-robust standard errors in parentheses. $N = 60$ cities, 19 countries throughout. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ 45
- 6 Heterogeneity by corpus size. The sample is split by tercile of $\log(\text{papers} + \text{projects})$: Small ($n = 21$), Medium ($n = 19$), Large ($n = 20$). Column (4) pools all 60 cities. Column (5) adds $\log(\text{cl. papers})^2$ to test for a non-linear depth–alignment relationship. HC3-robust standard errors. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ 46
- 7 Geographic robustness. Each column restricts the sample by country group. OECD membership follows the 2024 member list. All specifications include iGEM year span and first iGEM year as additional controls. HC3-robust standard errors in parentheses. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ 47
- 8 Carbon-capture intensity as a moderator. Dependent variable: city-level cluster overlap. Carbon-capture share measures the fraction of a city’s clustered papers (or projects) falling in the carbon-capture topic clusters. All continuous regressors are mean-centred; interaction rows show $\log(\text{cl. papers}) \times \text{CC share}$. HC3-robust standard errors. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ 47

9	Where cyanobacterial carbon capture has taken root. Cities with the most artifacts in the carbon-capture clusters (7, Cyanobacterial Metabolic Engineering; 8, Cyanobacterial Chassis Development; 62, Industrial Fermentation), counting projects, papers, and patents together.	57
---	---	----

List of Appendices

Appendix A	Custom Claude Code skills (discussion and frankGPT)	64
Appendix B	OpenAlex keyword query	67

List of Abbreviations

BBa	BioBrick registry part identifier (naming prefix)
BERT	Bidirectional Encoder Representations from Transformers
CC	carbon capture
DiD	difference-in-differences
DNA	deoxyribonucleic acid
FE	fixed effect(s)
HC3	heteroskedasticity-consistent standard errors (type 3)
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
iGEM	International Genetically Engineered Machine
NLP	natural language processing
OECD	Organisation for Economic Co-operation and Development
OLS	ordinary least squares
SPECTER2	Scientific Paper Embeddings using Citation-informed Transformers (v2)
SynBio	synthetic biology
UMAP	Uniform Manifold Approximation and Projection
USPTO	United States Patent and Trademark Office

List of Symbols

c	a city; the unit of analysis
P_c, Q_c	the sets of city c 's papers and projects
$\mathbf{e}_i$	embedding vector of document i , with $\mathbf{e}_i \in \mathbb{R}^{768}$
$\bar{\mathbf{p}}_c, \bar{\mathbf{q}}_c$	paper and project centroids (mean embeddings) of city c
$\bar{\delta}$	mean project-level alignment (own-city minus other-city similarity)
O_c	Jaccard topic overlap for city c
$N_{\text{proj},c}, N_{\text{paper},c}$	project and paper counts in city c
γ_k	fixed effect for country k
$\mathbb{I}[\cdot]$	indicator function (1 if the condition holds, 0 otherwise)
$Y_{c,k,t}, X_{c,k,t}$	downstream and upstream shares of cluster k in city c , period t
α_c	city fixed effect
$\delta_{k,t}$	topic-by-period fixed effect
$\beta, \hat{\beta}$	difference-in-differences coefficient and its estimate
r_M	Mantel correlation coefficient
$\hat{\Delta}$	permutation excess: observed mean overlap minus the null mean
B	number of permutation (or bootstrap) replicates
R^2	coefficient of determination
p	p-value

1 Declaration of Authorship

I hereby declare that the thesis submitted is my own unaided work. All direct or indirect sources used are acknowledged as references. I am aware that the thesis in digital form can be examined for the use of unauthorized aid and in order to determine whether the thesis as a whole or parts incorporated in it may be deemed as plagiarism. For the comparison of my work with existing sources I agree that it shall be entered in a database where it shall also remain after examination, to enable comparison with future theses submitted. Further rights of reproduction and usage, however, are not granted here. This paper was not previously presented to another examination board and has not been published.

2 Statement on AI use

AI tools were used over the course of this project. Claude Code, Anthropic's agentic coding assistant, was the primary generative tool used; it ran literature searches, processed and cleaned data, generated the language embeddings the analysis rests on, wrote and debugged code, reasoned about methodology, reviewed and constructed drafts, and collated notes. The full project's codebase lives in a public GitHub repository at <https://github.com/Zer0Juice/synbiomap>. Commits were made by the agent at regular intervals throughout development. The repository's history is a complete audit trail. Every script, notebook, and draft has a timestamped record of what was added, what changed, and why. Anyone who wants to inspect how this project was built, or replicate any part of it, can read it there.

I have chosen to document this use in full rather than minimize it. There is a stigma around AI in academic work, and while much of it is valid and I do have serious concerns over the growing role of generative AI in society, academics are currently incentivized to avoid publicly discussing use of AI. The solution to this period of change lies in openness, disclosure, and education instead of concealment and shame. I believe this project is much deeper and wider in scope than it would have been without the use of AI agents, which were extraordinarily useful in building and maintaining complex bodies of code. The web crawling scripts required to collect the data this analysis is built on would not have been tenable for me to build and run without use of these tools. Natural language document embeddings form a core part of the calculations used throughout this paper.

To this project the most valuable and experimental use of Claude Code was simulating debates or feedback. This was done through a series of extensively documented custom prompts (or Claude "skills"). These tools are simply markdown files: plain-text documents that set the project's goals, scope, and voice, and that steered the agents toward my intent. Two of them drove the reasoning work directly. A `discussion` command staged a debate among several named, real researchers, which pushed the model to draw on those scholars' actual bodies of literature rather than a generic average of the field. A `frankGPT` command cast the model as an economic-geography advisor in the mold of Frank Neffke, to pressure-test the framing and the empirical strategy. These files are reproduced in the appendix. They are the clearest picture of where the direction came from, worth reading alongside the results, and offer some level of informative entertainment. Of course, all prompt files are available via the GitHub repo linked above.

3 Introduction

For most of human history, the natural world has seemed like a fixed backdrop to our existence, a force as inevitable as it is mysterious, but this old model no longer checks out. As our technological abilities and knowledge of ecology have deepened, it has become very clear that human processes are shaping the biosphere at a momentous scale; we've reached a stage where human activity is the dominant driver of change in Earth's natural systems, so much so that scientists are proposing we have entered a new geological epoch: the Anthropocene (?). Whether we set out to or not, we are terraforming our own planet, and the current trajectory does not bode well. Rapidly shifting climate cycles, ocean acidification, collapsing habitats, a thinning web of life; this environment is no longer simply a backdrop we live in front of. Our world is a complex adaptive system we reside inside, molded by our actions whether intentional or otherwise.

There is, however, a tough shard of hope buried in that pile of dread and alarm. The scale of the damage can also be taken as a measure of our power and ability. A species that can warm a planet by accident, with no coordination and no shared intent, has proven something about its reach. If our collective action can disrupt the natural world this much, then our collective action might move it back. This is the thesis behind this research: the power to reshape the biosphere is in our hands. We simply must direct this capability down the right pathways.

We have always tried to understand the living world by taking it apart. For most of its history, biology was a descriptive science, breaking life into organs, then cells, then the genes beneath them. From those pieces, over time, an account of the rules that hold life forms together was developed. Biology had learned to read the language of life, yet still we lacked the ability to write it. That too is changing. Once the rules were understood well enough, biologists began to compose with them. A genetic sequence can be treated as a function written in the programming language of life, standardized, and snapped together with others into genetic circuits (?). Synthetic biology is descriptive biology picking up the pen. Of course, manipulating natural systems without fully understanding them is not without its risks. The same command over living systems that lets us read and rewrite a cell can be put to a range of uses. Cells can be engineered to make fuels without a drill, or to pull carbon back out of the air. They can be used as weapons, or as food. Carbon capture by precision fermentation is an exciting example highlighted throughout this project, where engineered microbes consume carbon dioxide and turn it into something useful. This is not a thought experiment; the technology has been built and implemented at industrial scale. Synthetic biology is one of the most potentially useful tools in our kit for intentionally saving the biosphere.

How that tool gets used is not decided in the lab alone. Synthetic biology is becoming a deeply influential industry, with associated opportunities, risks, and the inevitable regulations that thus emerge. Governments fund research, either directly or through academic institutions, writing rules that permit or forbid various technologies and shaping the markets that receive new innovation. Those choices carry real weight, impacting which problems get worked on and the

level of time and money it takes to bring a product to market. A field with this much potential for transformation is also a place where public policy has vital leverage. The decisions made now while the industry is still young and its norms are still forming will shape what it eventually grows into.

Making proper decisions about an emerging field depends on being able to see it clearly. After all, policymakers cannot fund what they cannot find. Understanding how a young field is growing and how its ideas move from one place to another is the practical case for studying innovation in the first place; If we want public investment in synthetic biology to have a meaningful impact, we need some way to trace the field's shape as it forms, rather than reading it off after the fact from a list of finished products. That means measuring research, the raw material of innovation, at a scale where conclusions can be drawn and informed decisions made.

Synthetic biology is unusually difficult to measure and observe clearly from its exterior due to its interdisciplinarity. It grew out of molecular biology, and their vocabularies overlap to a significant degree; a researcher studying the metabolic pathways of *E. coli* may have no intention of engineering them, yet would use the same words as a synthetic biologist who does. Oldham and colleagues confronted exactly this problem when building systematic academic and patent maps of the field (?). Synthetic biology can be readily identified from descriptive molecular biology, but it shares so much surface vocabulary with the broader literature that any keyword query pulls in a large number of papers and patents unrelated to engineering biology.

Keyword drift is another source of the problem. Synthetic biology names its new methods and organisms as it discovers them, and today's next big thing may be last year's unnamed result in different terms. For instance, CRISPR, perhaps the most transformative technology for genetics in recent history, was a known if obscure acronym long before it became synonymous with gene editing. If we were to run a keyword query built in 2010, it would miss the root of nearly all the modern genome-editing literature. The same problem applies to specific organisms and parts; cyanobacteria appear in the literature under many genus names and common names depending on the decade and the subfield. Any boundaries for synthetic biology calibrated in 2005 may have drifted by 2015. Drawing a clean and stable line around the field, and retrieving the right corpus of documents, is a genuine technical and conceptual problem that we grapple with throughout the data chapter.

Synthetic biology also comes with a data source that most young fields lack, at the place where the origins of this inquiry begin. The International Genetically Engineered Machine competition, iGEM, is a worldwide student contest in the field, somewhere between a science fair and a startup accelerator. Teams of students, usually hosted by a university, spend most of a year designing and building an original project from a standardized toolkit of synthetic dna fragments. Every team starts from the same set of interchangeable genetic parts and the same basic rules, but they craft a widely diverse range of projects. These are richly documented in a standardized, open source format on designated documentation websites called "wikis" (?). For many, this is their introduction to the field and a launching point into a career in academia or

industry. By collecting thousands of these projects, we build a detailed and open record of a field actively composing itself.

To empirically compare a student project against an academic paper or a patent, we need a way to relate documents across fields. We use a natural language model, SPECTER2 from the Allen Institute for AI, to process the titles and abstracts of a documents into vectors that encodes hidden layers of meaning (?). When we represent these vectors as points in a high-dimensional space, documents about similar things land near one another, whatever words they happen to use, so the distance between two vectors becomes a measure of how related two pieces of work are. SPECTER2 was trained to predict if papers cite one another, and we use semantic distance as a proxy for the kind of relatedness a citation would mark. It is the core beneath every comparison in this thesis, and the reason we can compare patents, projects, and papers so readily.

The question this we ask is simple. Do cities whose students research certain areas in synthetic biology also produce academic papers, and patents in related areas? Is iGEM research embedded in the topical landscape of a city's research? We arrive at a qualified yes. Measured carefully, a city's student projects and its academic papers tend to fall in the same specific topics. Patents line up topically with papers, and papers line up with projects, though patents and projects remain further in semantic space. We approach this as relatedness and co-location, not interpreting anything as proof of causation.

Prior work on innovation geography measures relatedness using labor flows, product similarity, and co-patent networks, but those methods are designed for large, mature industries and require data on worker mobility or product co-export that simply do not exist for a student-academia-industry pipeline in an emerging field (??). Prior work on iGEM has studied team collaboration and career outcomes, but has not connected projects with other products of synbio research. We close this gap by fine-tuning a model on links between the artifact types, building a shared embedding space on a brand-new dat set, and testing the result against an independent validation from physical genetic data the model never touched.

To refine the scope, we focus on carbon capture technology, identifying clusters in the semantic space and exploring them more in-depth. This is one of the most potentially transformative technologies Synthetic Biology could help to build and scale. Cyanobacteria, the blue-green algae, are among the oldest photosynthesizers on the planet, and they earn their place here twice over. They are a favorite chassis for engineered carbon capture, cells that eat sunlight and carbon dioxide and can be tuned to generate fuels or other useful products in turn. They are also a topic that unites iGEM students, academic labs, and patent-filing firms in the same cities. Following this thread through the data allows us to zoom in on a particular case and see the industry from a different perspective. The core focus of this thesis is about relatedness across a field, but algae are where we watch it happen.

Underneath the embeddings and regressions, the data tells a simple story. Tools for synthetic biology are open. Students learning to use them are doing so, alongside local labs and firms working on the same problems. Knowledge is locally shared across sectors of the industry.

Where those students cluster is a signal of where a field is putting down roots. If that signal is real, then funding student science and the open research that sustains it is one of the surest early investments a government can make in an emergent field.

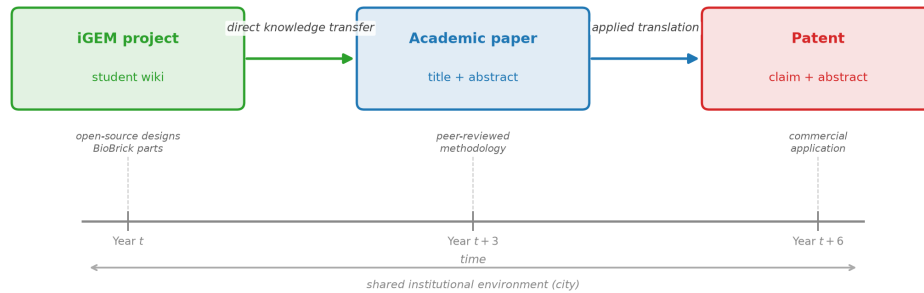


Figure 1: Conceptual model of the three-artifact innovation ladder. Student iGEM projects introduce early-stage biological designs at time t ; academic papers develop and disseminate those ideas roughly three years later. patents mark downstream commercial translation at a further lag. All three activities co-occur within a shared urban institutional environment whose role is the central object of study. Arrows label the proposed mechanism; the sequence is the hypothesis this thesis tests.

4 Background

4.1 Innovation, research, and the role of the state in the Bioeconomy

Innovation policy depends on the simple concept that money spent on research today pays in the form of capability tomorrow. This idea is not new; Vannevar Bush made the canonical case for it in 1945, arguing that basic science funded by the public and pursued without a fixed application in mind is the seedbed from which practical advances eventually grow (?). This framing has shaped decades of science funding, with the expectation that with support for research, breakthroughs and applications are soon to follow. While this model isn't wrong, it does happen to be incomplete; it treats the flow from discovery to application as an automatic pipeline, freely running on its own once the money enters as input and outputting influential research.

The reality is messier. The pipeline model, with research at one end and products at the other, misses how much the two shape each other, and how much the public sector impacts the private. Governments act both as funders and active participants in innovation. The public sector plays the role of early customers and collaborators that help new technologies cross from lab to market (?). Whether a young technology survives the gap between a working prototype and a viable product often depends on public support. Innovation studies has spent decades documenting how much of this is collaborative and institutional, depending on the surrounding environment

rather than on lone innovative genius.

Innovation is built based on the environment around a discovery, and the public sector plays a vital role in creating it. They build the conditions in which research can emerge into capability by funding education and shared infrastructure. Standards must be set that allow for healthy competition, while governments keep an eye on potentially transformative fields so they can act as early partner rather than late regulator. In synthetic biology, where products impact public health, food, energy, and the environment, the state will be vital in determining where new capabilities emerge. Acting as an early partner requires understanding the field, and emerging fields are the hardest to truly know. A government that wants to support synthetic biology has to know where it is happening, who is doing it, and what counts as part of it. However, none of these factors is obvious for a field where boundaries are still forming. Usual instruments will lag behind, with industry classifications and funding categories built for established sectors missing the true innovation that slips under the radar. Technology that does not fit existing boxes is easy to undercount or miss, but the communication gap runs both ways. Researchers struggle to convey the shape of a moving field to the people who fund it, and funders struggle to conceptualize what they are asked to support.

The difficulty is worth the trouble, because new fields are where the largest gains in productivity tend to emerge. New technologies open the most room for growth precisely when they are young and unformed, before their methods settle and their returns level off. This is the frustrating shape of the problem for policy: the fields with the most to gain from good early decisions are the ones we can see least clearly. A synthetic biology that is easy to measure will be a mature synthetic biology, with its trajectory mostly set. If public investment is going to shape the field, it has to act while the picture is still blurry. This highlights the potential importance in any method that can bring the emerging picture of such a field into slightly better focus.

If research is the raw material of innovation, where does it come from and how does it accumulate? It's not random, and it's not evenly distributed; over time, it's natural for certain places to develop an advantage over others. A place, a firm, or a field is most likely to move into activities related to what it already does, because the skills, and knowledge carry over easily (?). Economic Geography has helped in exploring the differences between regions' innovative potential: why different areas move to specialize in different innovative technologies. As cities develop their capabilities in new industries and products, they tend to specialize in certain technologies over others. This could be due to geography (for instance, a city near a body of water will be better equipped to engage in shipping than a landlocked peer located in a desert), demographics (areas with younger populations have a greater advantage in industries requiring manual labor), or a wide range of other factors. As cities specialize in a technology, they continue developing their comparative advantage in this and related areas, branching out into a range of similar innovative technologies (?) . In the case of SynBio, this would mean that a firm developing capabilities in, for example, scaling industrial fermentation of ethanol would mean enhancing a city's potential to develop capabilities in production of other biofuels.

4.2 A field worth watching

Synthetic biology is a field where these cycles of innovation, these economic theories around the formation of new industries and technological shocks play out in fast motion. Two cost curves drive it. DNA sequencing has fallen in price faster than computing's own famous Moore's Law decline: the human genome cost roughly three billion dollars to sequence in 2003 and a few hundred dollars today, a drop of seven orders of magnitude in just two decades. The cost of DNA synthesis, writing a custom genetic sequence from scratch, has followed, turning what was once a prohibitively expensive operation into a routine service that any well-equipped lab can order online. As sequencing and synthesis get cheaper, the barrier to designing and testing new biological systems falls, and more researchers enter the field with more hypotheses to try. The result is a field whose rate of change compounds on itself: each year's discoveries lower the cost of the next year's experiments, and the methods change faster than any single literature search can follow. What Drew Endy described in 2005 as the "foundations" of a new engineering discipline (?), design-build-test cycles using standard interchangeable parts, has since been built out into an industry with commercial gene synthesis companies, automated laboratory platforms, and machine-learning tools that design proteins from scratch. The field is still developing, making it both unusually hard to study and unusually important to track.

The stakes match the speed. The same capability that lets a lab engineer a microbe to capture carbon or produce a medicine could, in other hands, be turned toward harm. These tools have the potential to create incredibly dangerous and destructive bioweapons, or to alter our natural world in fundamental and irreversible ways. While this paper tends towards an optimistic view of this developing industry, the dual-use worry is very real and widely discussed across the field (?). This is a technology with a genuinely enormous potential to help, deeply real potential to harm, and a trajectory that is still evolving. A field like that should not be left unstudied. Understanding where it is going, and steering it while steering is still possible, is part of our responsibility to engage with technology in a safe and productive manner.

How, then, do we watch and measure a field like this? We read and analyze its outputs. Patents are an obvious place to start when studying new technology; they are, after all, filed because someone believed an idea had enough commercial value to be worth protecting. Patents have long served as a window on the rate and direction of innovation, and a large literature uses them to track how science turns into technology, including the growing extent to which patents cite scientific papers (??). A patent tells us that an idea reached the point of expected commercial use, where it was filed, and roughly when. For a field with industrial impact, this is a signal worth collecting.

The academic paper is the second output type, and it measures innovation from a different perspective. While a patent marks expected commercial value, a paper marks a bid for scientific impact; its authors are claiming a contribution to what the field knows, and the paper's reception measures whether the field agreed. The two signals often diverge. A breakthrough that reshapes

a science can have little commercial use, and a lucrative product can rest on an idea that means little to researchers. That divergence is useful. Patents and papers are partial reflections of the same underlying activity, and reading them together gives a fuller picture of a place's capabilities than either does alone.

4.3 iGEM Projects

There is another artifact type that makes synthetic biology unusually tractable to study: the student iGEM project. Each team's project is incredibly well documented in a standardized fashion, with much more fine-grained data available than comes with the average publication. They document problems, an approach, methods, experiments, models, results, collaborations, and contributions to the iGEM Registry, tagged with the team's institution and year. The competition runs a public registry of these wikis and of the genetic parts teams create, so the entire archive of iGEM history is accessible and consistently structured. The competition has produced thousands of teams across more than 40 countries, with every team record including location, year and generally a usable abstract. There is no other resource in any scientific field that provides this kind of systematic, open, annual geotagged record of student research at scale (?).

A student project carries yet another different kind of signal from a patent or a paper. It records what a group of undergraduates, guided by local mentors, chose to work on with one year and a shared toolkit, without the same level of commercial stake that shapes a patent and reputational pressure that shapes a peer-reviewed article. Both of those stakes constrain what gets filed or published; a student project is shaped mainly by the mentors' research interests, the students' curiosity, and the immense pressure of the competition structure itself, which compresses the timeline of research into a standardized timeline. This shows a different filter into what local research environment cares about, what ideas might be floating around in the air. Where a city's students point their projects is a third view of the city's technical orientation, and the one closest to the beginning of a research career, before the selection pressures of publication and commercialization have started to work their magic.

4.4 BioBricks and the standardized parts registry

iGEM's shared toolkit is a public registry of standardized DNA sequences called BioBricks. A BioBrick is a small piece of genetic function with a documented job, like a promoter that switches a downstream gene on, packaged so it snaps together with any other part. Teams deposit the parts they build and draw on the parts every earlier team left behind. This is the LEGO logic of synthetic biology made literal: standard bricks with known behavior, open for anyone to reuse. It is also what turns a yearly competition into something cumulative. Each generation of projects adds to a catalog the next one inherits, so a team today begins where last year's teams stopped instead of starting again from raw DNA.

The idea came from Tom Knight at MIT in the early 2000s, who wanted to engineer biology

the way we engineer electronics (?). An engineer choosing a resistor from a catalog does not ask how it was made; she reads its rating and trusts it to behave. Knight wanted a genetic part to carry the same kind of guarantee, so that building a living circuit could become a matter of choosing components whose behavior was already documented. The standard that made this possible was published as a community Request for Comments, the same open document format the internet's protocols are written in, and it specified the flanking DNA that lets any two compatible parts be joined in a single reaction (?). The format was itself an argument: a standard for engineering life should be published in the open and owned by its community, the way the protocols that run the internet are.

The Registry of Standard Biological Parts opened with the first iGEM competition in 2004. Every team is expected to leave at least one characterized part behind, so the registry has grown in step with the competition, from a few dozen parts that first year to roughly 89,000 by the time we downloaded it. Each entry holds the DNA sequence, a functional description, a type, the depositing team, and often a pointer to the paper the sequence came from. Composite parts, the ones built by joining simpler parts together, list their components, so the catalog doubles as a map of how small parts combine into larger ones. Read a single composite entry and you can follow it down through its pieces to the original sequences it was assembled from.

Assembly itself has moved on since Knight's first standard. The restriction-enzyme ligation it was built around is slow and prone to error, and for anything complicated it has given way to faster methods like Gibson assembly and Golden Gate cloning. The chemistry changed. The identifier did not. Teams still deposit in the standard format and still cite registry parts by their BBa_ numbers in their papers and project wikis, so a BBa_ code has become a stable handle for a piece of biological function, whatever technique actually stitched it into a given construct.

This is what makes the registry useful to us as more than history. The parts a city's teams deposit are a physical record of what those students set out to build, and that record owes nothing to the words they wrote about the work. A team chasing biosensors leaves behind reporters and inducible promoters; a team engineering a metabolic pathway leaves composite parts that chain enzymes end to end. The registry knows nothing about embeddings or topic clusters. It records only the parts themselves, and that independence is exactly what lets it check our work: if the topics our model reads out of a city's project wikis match the parts that city's students actually deposited, the agreement cannot be an artifact of how a model reads an abstract. It has to come from the biology. The Mantel test in the results chapter is built on this logic.

4.5 Booze Bugs

To see what these projects look like, and how they connect to the research around them, it helps to follow one. In 2009 the Uppsala iGEM team built a project called *Booze Bugs: Sun to Alcohol*, one of many teams that have worked on creating a sustainable fuel source. Conventional biofuel is fermented from starch and sugar, which puts fuel crops in direct competition with food; the Uppsala students wanted to skip the crop entirely. They turned to *Synechocystis*, a

cyanobacterium, one of the blue-green algae that have been running photosynthesis for billions of years, and engineered it to convert sunlight and carbon dioxide straight into ethanol and butanol. A microbe that drinks sunlight, eats carbon dioxide, and exhales fuel is a pure statement of what synthetic biology can promise, and when these sorts of technology are successfully scaled up to industrial levels they can prove incredibly transformative.

The impact of the research did not end with the competition. Two supervisors of the Uppsala team, Daniel Camsund and Thorsten Heidorn, continued in cyanobacterial research in Peter Lindblad's lab at the same university, and in 2014 the three published a paper on engineered promoters for cyanobacteria in the *Journal of Biological Engineering* (?). In this paper, they cited a collection of genetic parts from the iGEM registry by their BBa_ identifiers, including one deposited by a former teammate. The Uppsala team contributed one called BBa_K273019, a copper-inducible promoter: a genetic switch that turns a downstream gene on in the presence of copper, tuned to work inside their cyanobacterium. Five years after it was deposited by undergraduates for a competition, the same sequence appears as a cited building block in the peer-reviewed literature. That is the kind of link this project was built to trace.

Booze Bugs points at something bigger than one team. The organism at its center, cyanobacteria, is one of the most consequential life forms in the planet's history, and one of the most promising tools for its near future. To see why the algae matter so much, and why so many teams, labs, and companies keep returning to them, it is worth stepping back from Uppsala to the algae themselves: what they did to the planet once before, and what we are now trying to get them to do on purpose.

4.6 Algae, oxygen, and carbon capture

Cyanobacteria have reshaped the planet before, and catastrophically. For the first couple of billion years of life on Earth, the atmosphere held almost no free oxygen, and the living world was built to do without it. Then cyanobacteria invented oxygen-producing photosynthesis and slowly filled the air with a gas that was poison to most of what was alive. The result, around 2.4 billion years ago, was the Great Oxidation Event: a wholesale remaking of the atmosphere that drove countless anaerobic organisms to extinction, often called the planet's first great mass die-off.

If the Great Oxidation Event is what cyanobacteria did to the planet by accident, fermentation is the oldest example of us putting microbes to work on purpose. Human control over fermentation predates written history; people were brewing beer in China by around 7000 BCE, long before anyone knew a microbe existed.

This is no longer an idea that lives only in the laboratory. Different lineages of research have pushed these organisms toward industrial use, and are worth distinguishing because they map closely onto clusters described in the data chapter

The first runs directly through cyanobacteria. Algenol, founded in 2006 and based in Fort Lauderdale, Florida, engineered cyanobacteria to secrete ethanol directly into the growth medium,

bypassing the harvest and extraction steps that make algal biofuels expensive, similar to the Uppsala students. Joule Unlimited, a startup founded in 2007 out of Cambridge, Massachusetts, used cyanobacterial fermentation to produce a biodiesel substitute with no biomass harvest required. Photanol, a spinout from the University of Amsterdam, followed the same logic with organic acids and alcohols as target products, taking an explicitly academic-to-commercial pathway in which the founding science came directly from published research on cyanobacterial metabolism. Each of these firms appears, at least in part, in the USPTO patent record: their inventors file in the United States to protect commercially valuable inventions, and those filings are part of the patent corpus in this study. The cities they cluster in, Cambridge, Fort Lauderdale, Amsterdam, and a handful of others, are consistent with the cities active in the clusters identified and explored in the results chapter.

The second lineage uses a different organism but the same closed-loop logic. LanzaTech, founded in New Zealand and now headquartered in Skokie, Illinois, ferments carbon-rich industrial waste gases, steel-mill exhaust, refinery off-gas, municipal solid waste, into ethanol and other chemicals using acetogenic bacteria. The organism is not cyanobacterial, and the energy source is not sunlight but the chemical energy in industrial off-gases. But the conceptual move is the same as in the Uppsala project: take a carbon stream that would otherwise be emitted into the atmosphere, feed it to an engineered microbe, and recover something useful in return. LanzaTech's technology has reached commercial scale, with plants operating at steel mills in China, India, and Europe and a collaboration with Virgin Atlantic to produce sustainable aviation fuel. The New Zealand origin of the company, and the strong USPTO patent record that followed its move to the United States, is precisely the kind of trajectory the project-to-patent path in this thesis is designed to trace: an idea that began in academic and industrial research in one country, developed through patented stages in another, and eventually scaled to global deployment.

The shared premise across both lineages is the one the Uppsala students were chasing in 2009: instead of pulling carbon out of the ground to make a fuel and burning it back into the air, close the loop and grow the fuel from carbon already present in the atmosphere or in a waste stream. Carbon capture by fermentation has moved from proof of concept to running commercial plants within roughly two decades of the foundational academic work. It is the clearest existing answer to the question the introduction posed: whether the same biological power that damaged the biosphere can be aimed at repairing it. The speed of that transition, from student competition project to industrial-scale plant in fifteen years, also makes it a compelling case study for the thesis's central question about how local innovation ecosystems form and what role student science plays in them.

4.7 What iGEM leaves behind

Uppsala's thread from project to paper is one of many kinds of afterlife an iGEM project can have. Some become companies. iGEM has grown a dedicated entrepreneurship track around exactly this pattern, and its alumni have gone on to found synthetic-biology startups.

Others become papers, as the Uppsala case showed, and the registry leaves a documented trail of this. When we searched the full text of the open-access literature for iGEM part identifiers, we found hundreds of published papers that cite BioBricks from the registry by name, a lower bound on a much larger real influence. These are papers whose authors reached into the students' shared toolkit and used a specific part. The link runs the other way too: many BioBricks are themselves built directly from published research, translated into standard form. Project and paper are not separate worlds; they borrow from each other continuously.

Put these afterlives together and a picture forms of iGEM as an entry point into the whole system. A project can become a company, a paper, a part that other researchers reuse, or a career, and often several at once. The same competition feeds the commercial, academic, and open-source sides of synthetic biology from a single starting point, and it does so in the open, year after year, tagged by place and time. Few other corners of any field are documented this thoroughly this early in a researcher's life.

That openness is not incidental; it is the point, and an ethic the field takes seriously. iGEM is built on shared standards and public documentation: the parts are deposited for anyone to use, the projects are published as open wikis, and the norms of the competition reward teams for contributing back to the commons. This is the same spirit that runs through open-access publishing and public data, and it is what makes a study like this one possible at all. Knowledge meant to be shared and built on, like the BioBricks themselves, leaves a record that others can read.

So there is a strong case for studying iGEM directly, as a window on how synthetic biology forms. The students in the competition are the field's next generation of researchers, founders, and inventors, learning the standards and choosing the problems their later careers will build on. If we want to understand where the field is heading, watching where it is being taught and practiced by newcomers is a reasonable place to look. The students are an early draft of the field, not a sideshow to it.

For all that, the link between iGEM and the surrounding research landscape is underexplored. The competition is well known, but treating its projects as data, and comparing them to academic and commercial output, is rare. The main body of quantitative work on iGEM comes from Marc Santolini and colleagues, who study the competition as a model system for team science, tracing how teams collaborate and how their networks shape what they produce (?). That work establishes iGEM as a serious dataset. It leaves open the question this thesis takes up: how the projects relate to the papers and patents in the same cities.

The dataset rewards the attention. Each project comes with a place, a year, a team roster, an abstract, and a set of parts linked to other parts and to the literature, which together form a connected record of a field growing in the open. It invites a long list of questions. Do iGEM alumni cluster in the firms and labs that later hire them? Do teams cite researchers from their own city more than chance would predict? Do the parts a team builds forecast what the city patents a decade on? This thesis takes only one of these questions and pursues it carefully. The

rest are left standing, because a dataset this rich deserves more than one pass.

4.8 Why might local alignment exist?

Before measuring whether a city's student projects and academic papers are topically aligned, we should explore some possible mechanisms for this alignment. Three mechanisms are outlined, and distinguishing between them matters for interpretation of results

The first is direct knowledge transfer. Faculty members who supervise iGEM teams also run research labs; in many cases the iGEM advisor is a PI whose publications are in the same database as the papers we compare against. If a mentor's research agenda sets the direction of their team's project, alignment follows mechanically from the advising relationship. This channel is person-level: the transfer runs through an individual who appears in both registers, predicting that project topics should resemble the specific research interests of the faculty advisor, that alignment should be strongest in cities with large, research-active universities where advisors have deep publication records, and that student projects should precede any new papers the advisor writes on that theme by a few years, since the project is usually early-stage relative to a published result.

The second is a shared information environment. Perhaps a city's students and its academics converge on the same topics not because they influence each other directly, but because they respond to the same stimuli. Funding priorities, the same institutional research focus, visiting seminars, and local firms whose problems become projects and papers simultaneously can all influence research priorities. This is a common-cause mechanism. Neither group needs to observe the other; both are shaped by the same environment. Under this mechanism, alignment might show up simultaneously across artifact types, with no preferred direction, and the lead-lag curve should peak near zero. It should also survive the departure of any individual mentor: if the environment drives the alignment, losing one advisor does not break the city's profile.

The third is selection. Students choose which institution to join partly based on what the institution researches, and students with an existing interest in cyanobacteria may seek out Uppsala precisely because of its academic profile in that area. If sorting dominates, the alignment reflects student preferences meeting institutional offerings, not a property of how knowledge develops locally. This mechanism predicts alignment that is present from the start of a city's iGEM history and does not grow as the ecosystem matures, since it is driven by individual matching rather than cumulative spillover. It also predicts that the lead-lag structure should be flat: sorted students do not precede the papers; they are already at the institution that produces them.

These three mechanisms are not mutually exclusive, and the data cannot cleanly identify which one dominates. What they share, and what makes the project-paper alignment meaningful regardless of which mechanism produces it, is the implication of a substantive connection between student and academic synthetic biology at the city level. The direct-transfer and shared-environment mechanisms both describe how knowledge is organized locally; the selection

mechanism describes how capable people are allocated to places. In all three cases, a city where iGEM teams and academic labs work on the same specific topics is a city whose synthetic-biology ecosystem is coherent in a way that is unlikely to be manufactured by data artifacts. The robustness analyses in the results chapter are designed to rule out data artifacts, while the mechanisms describe what remains once they do. We stand up our assertion, then attempt to knock it down with every tool at our disposal.

4.9 The geography of innovation

Economic geography turns relatedness into a research program with real predictive power. The principle of relatedness holds that regions diversify into industries and technologies related to their current strengths, and the pattern is regular enough to forecast: knowing what a place specializes in today predicts what it will move into tomorrow (??). The same logic has been carried from products, to patents, to scientific fields, and to more complex, high-value activities that regions compete to attract (?). Relatedness is the through-line, explaining why capabilities cluster and spread to some places and not others, why a region's past shapes its future.

So innovation is not geographically neutral; knowledge does not flow frictionlessly to wherever it would be most useful. Instead, it stays sticky, concentrated near the people and institutions that produced it. Knowledge spillovers, the informal spread of ideas from one worker or lab to the next, are strongly local, fading with distance (?). Part of the reason is that the most valuable knowledge is tacit, learned by doing and passed hand to hand, master to apprentice. A discovery made in one city is more likely to seed a related advance in the same city than a similar one across the world. Geography shapes where innovation goes next. Since this is such a well established effect, it might be more of an interesting result to show that iGEM research *isn't* geographically influenced; that would show that the vision of a truly **International** Genetically Engineered Machine had actualized, that Synthetic Biology had transcended borders and become a global collective.

But *how* do capabilities and knowledge cluster spatially? The answer is grounded in economic literature; Duranton and Puga trace agglomeration to three micro-level mechanisms that operate at the scale of workers and firms rather than regions: sharing, matching, and learning (?). Sharing of knowledge happens when firms that concentrate together can sustain more specialized suppliers than any one firm could support alone, reducing input costs for all. Matching is more efficient in thicker markets: a researcher searching for a collaborator with a very specific skill finds one much faster in a city of ten thousand researchers than in a town of fifty. And learning is faster when dense co-location brings people into incidental contact. The conversation at a conference, the seminar attended on a neighbour's recommendation, the former colleague now at a competing lab; these sorts of mechanisms compound when the people and firms in question work in related areas. A shared vocabulary makes learning cheaper, overlapping skill sets make matching more likely, and complementary inputs make sharing more valuable. These are the micro-foundations of what the relatedness literature documents at grander scales.

Alfred Marshall identifies a complimentary observation from three sources of spatial concentration: shared input markets, pooled labor, and knowledge spillovers (?). These can be mapped onto iGEM; the shared input is the BioBrick registry, an open-source toolkit that all teams draw from and contribute to, a commons reducing the cost of working on the same subfield regardless of location. Pooled labor is the graduate student and postdoc population that circulates between student teams and academic labs, the same people who learn a method on an iGEM team and apply it in a PhD project. Knowledge spillovers are the face-to-face transmission of tacit knowledge, exactly what co-location in a city is supposed to enable. Hallway conversations transfer know-how that raw citation data cannot hope to capture (?). iGEM is unusual in that the relevant “labor market” is transient; students move through an institution for a fixed, short period, unlike most workers moving between firms. Much of the shared input is also openly licensed rather than commercially supplied, but the logic of agglomeration still applies: cities where multiple forms of synthetic-biology activity concentrate together are cities where the externalities compound. The Marshallian framing predicts exactly the kind of local topical specialization we set out to measure.

Jane Jacobs added a different kind of agglomeration mechanism particularly relevant for a young field like synthetic biology (?). Where Marshall’s externalities come from *specialization* within an industry, Jacobs’ externalities come from *diversity* across industries. Cities that mix many different activities generate cross-pollination between fields, sparking new combinations and innovations. Synthetic biology is itself a product of this kind of encounter, sitting at the intersection of molecular biology, computer science, and chemical engineering, which had themselves all emerged from different disciplines. Fields that are still defining their own boundaries benefit more from this melting pot of a diverse city than from the Marshallian depth of a highly specialized one, because they need to borrow tools and concepts from adjacent areas as fast as they can generates its own. The empirical prediction is that synthetic biology should cluster in large diverse cities rather than narrow specialist clusters, broadly consistent with the distribution of iGEM activity and academic output in this corpus.

If capabilities are local and path-dependent, then measuring them matters, but measuring them well is difficult. Capability and innovation are abstract; we only ever observe the traces, outputs a research process leaves behind. To study them at scale we need those traces in a quantitative form, standardized enough to compare across places and times. The choice of what to measure, and how, carries a tremendous amount of weight here; measures that captures the wrong thing will describe the world confidently and incorrectly. We explore a range of measures, with a range of correctness levels, further in the results section. Most of the methodological care in this project has gone into getting these measures to reflect real relatedness rather than an artifact of how the data is being manipulated.

The natural place to measure knowledge spillovers across firms and industries is the city. Innovation in deep-technology fields concentrates in cities; cities gather all the ingredients research needs into one place. Universities with equipment and students, firms with adjacent

expertise, investors willing to fund risk, and a dense labor market that lets people and ideas move between them. Boschma and colleagues study exactly this at the city level, tracing how scientific relatedness shapes which biotechnology capabilities a city develops over time (?). The city is small enough to be a coherent unit and large enough to contain a full innovation ecosystem, making it the right unit of analysis for comparing research across projects, papers, and patents.

Two further arguments support the city as the unit of analysis. The first is the geography of labor markets. Workers, students, and researchers circulate within cities far more than between them: a graduate student commutes between university and a collaborating lab; a postdoc moves between two institutions in the same metropolitan area; a founder recruits from the alumni pool of the nearby university. The city approximates the catchment area of this circulation, capturing the labor market in which capabilities pool and recombine (?). A coarser unit, like a national region or a country, would pool cities with very different specializations into one observation, blurring the within-country heterogeneity that is most interesting (it's also worth noting that in an industry dominated by a few large US and Chinese cities, country-level comparisons are quite distorted). A finer unit, like a neighbourhood or a campus, would miss the cross-institution dynamics that depend on city-level density. The second argument is practical: the city is a standard geographic unit used worldwide at which all three of our artifact types can be geocoded with reasonable reliability. Patent inventor addresses resolve to cities, while author affiliations and iGEM teams resolve to the cities of their host institutions.

4.10 Measuring relatedness

To ask whether a city's projects, papers, and patents are related, we need a way to measure relatedness that works across all three, and the three could hardly be more different on the surface. A student wiki, a journal abstract, and a patent claim are written by different people, for different readers, under different rules, in different registers. A method that measures relatedness between two papers may say nothing useful about a paper and a patent. Before any of the geographic questions can be answered, we need a yardstick that reads across artifact types, and fortunately Economic geography offers several candidates.

There are several methods to evaluate relatedness between industries or technologies, each with an array of strengths and weaknesses (?). Citation networks are a simple method of determining co-occurrence between artifacts such as papers or patents; if there is a citation linking two artifacts, they are related. Semantic overlap, measured as keyword co-occurrence as used in (?) is also widely used. Technological class co-occurrence is another simple measure that can be used to link patent filings. Revealed Co-Occurrence marks two entities as related if they show up together more often than chance would allow (?). Relatedness can also be calculated by resource flows between two entities; for instance, if one firm buys a product from another, or if employees from one regularly seek work at another (Neffke Henning 2012). However, each of these measures has drawbacks. Citations can reflect other biases and are often a targeted measure, leading to manipulation (?), while not being conducive to links between different artifact types.

Keyword co-occurrence loses some accuracy as fields and their associated vocabulary evolves over time, due to the concept of keyword drift. Technology classes only can measure relatedness between patents, and don't accurately measure emerging fields (like SynBio) (?). Revealed Co-Occurrence requires spaces for artifacts to co-occur *in*. We're aiming to measure relatedness across several different artifact text types in a fast-moving field, so we need a measure that's sector-agnostic. Keyword co-occurrence would be principled and established in the literature, but keyword drift could get in the way of a meaningful measure that holds up across decades of time-series data.

The measure this thesis arrives at is best understood against Boschma's taxonomy of proximity types (?). Boschma argues that productive interaction between actors requires proximity along at least some of five dimensions: cognitive, organizational, social, geographical, and institutional, with cognitive proximity closest to what we are measuring. This is the degree to which actors share a knowledge base and can therefore absorb and evaluate each other's work. Two groups with high cognitive proximity, working on the same technical problems with the same methods, are more likely to benefit from knowledge exchange than two groups that happen to share a postcode but work in different areas of the field. Our semantic topic overlap is a proxy for cognitive proximity revealed through documented output. If two cities' student and academic communities are cognitively proximate to the extent that they concentrate on the same specific research clusters. Geographical proximity, which is what co-location in a city provides, is a necessary condition for the face-to-face interaction that transmits tacit knowledge, but it is not sufficient for knowledge exchange without the cognitive proximity to use what is shared. Both are present, by construction, in the cities where iGEM teams and academic labs co-locate around the same topics. Students and academics share both a knowledge domain and a geographic location, which the agglomeration literature identifies as productive (?).

For this project, though, most of these measures do not fit, and understanding why clarifies exactly what we need.

The input-based measures, those built from shared labor or skills, need data on who is moving where. Labor flows between iGEM alumni and biotech firms or shared personnel between student teams and patent applicants might be interesting to study, but no such dataset exists at the scale or coverage this project needs. The few studies that have followed iGEM alumni into careers do so with small, hand-curated samples (?); a city-level analysis using labor flows would require a dataset that has not been built.

Revealed co-occurrence measures also need the same units to appear together many times over. The product-space method works because there are hundreds of countries, each exporting thousands of products, so the co-export frequencies are well estimated. A city's iGEM teams in a given year might be a handful; the few co-occurrence events between student projects and patents in the same city cannot be treated statistically in the same way. These measures were designed for economies producing large numbers of diverse activities simultaneously; a city producing five student projects and thirty patents in a given year is too sparse for revealed co-occurrence to

be reliable.

Finally, the content-based measures designed for a single artifact type, keyword overlap or citation networks, do not transfer cleanly across three very different kinds of document. A patent claim shares almost no vocabulary with a student wiki, and citation links between them are rare and biased. We need something that reads relatedness from what every artifact does have in common: a title, an abstract, a description in the vocabulary of synthetic biology, regardless of how that document was produced or what it was produced for.

The closest fit among the standard tools is textual. We must measure relatedness from the words the documents use, relying on the properties they have in common. Every project, paper, and patent has a title and an abstract, written in the vocabulary of the field, so keyword overlap is at least available across all three. Boschma and colleagues use exactly this kind of textual relatedness to study biotechnology at the city level (?). However, keyword matching is blunt, and misses that two documents can describe the same thing in different words. It stumbles precisely where synthetic biology is hardest, on a shifting vocabulary shared with molecular biology (?).

The probabilistic topic model is one step up from keyword overlap. Latent Dirichlet Allocation (?) represents each document as a mixture over a fixed number of latent topics, with each topic represented by a probability distribution over words. Documents about cyanobacteria and ethanol production land in similar topic mixtures even when they use different specific terms, because the underlying vocabulary distributions are closely related. This method is far more advanced than keyword counting, finding shared subject matter beneath surface word choice. It can also be applied to any document with text, including patents and student wikis. The problem is that the topics are fit separately for each document type or, if fit jointly, are dominated by the largest and most vocabulary-rich document type. In a corpus where papers outnumber projects and patents several times over, the latent topics are effectively paper topics discovered by a model that has seen far fewer patents and wikis. When a student project is mapped into those paper-derived topics, the assignment reflects how well the wiki's words fit the paper vocabulary, not how related the project is to the paper's subject. More fundamentally, topic models encode vocabulary distributions and nothing else; they cannot leverage the citation structure that tells us which documents humans have already judged to be related, and they cannot transfer knowledge from a large pre-trained corpus about the deeper semantic relationships that citation-trained models encode. For cross-type comparison in a field with heterogeneous writing styles, topic models solve the wrong problem.

Citations are the other content-based option, and they are appealing because they represent explicit, human-made links between two documents. They are strong evidence of relatedness when they exist, and a long tradition uses citation networks to trace the structure of science and its links to technology (?). Unfortunately, citations are sparse across the boundaries we care about. Student projects rarely cite each other in any standard form, patents cite papers unevenly, and the cross-type links that do exist, like the BioBrick citations we recovered, are a thin and biased sample of the true connections. Citations also carry their own distortions, shaped by

strategic and social pressures as much as by intellectual debt (?). As a primary measure across three artifact types, they leave too much of the graph empty, although they do form an important part of our model training data. Instead, we will use a newer, but still well-established, tool for calculating relatedness: semantic embeddings.

4.11 Semantic Space

We want to quantitatively compare text, but natural language conveys deep meanings that can be difficult to manipulate with mathematics.

To empirically determine whether two artifacts are related, we must turn their text into numbers, then compare them while preserving their underlying meaning. The traditional method of keyword co-occurrence runs into some problems in emerging fields like synthetic biology, where the vocabulary overlaps heavily with established fields such as descriptive molecular biology. A paper on engineering metabolic pathways can look, for instance, much like a paper that only describes those natural metabolic pathways with no interest in creating new ones (?). We need a representation that captures the true meaning of a text rather than just overlap of a few key terms.

The methods and concept we rely on emerge from linguistics and information science. Words appearing in the same contexts tend to mean similar things; for instance, "Ferment," "culture," and "anaerobic" regularly turn up near one another, while "ferment" and "bumblebee" do not.

By tracking the contexts a word appears in across a large enough body of text, context patterns begin to stand in for the word itself and reflect the underlying meaning contained in the word. Turn each pattern into a set of numbers, and words used in similar ways end up with similar sets. Semantic meaning becomes represented by location in a shared "semantic space".

The first practical versions of this were built through our old friend keyword co-occurrence; as linguist JR Firth, put it "you shall know a word by the company it keeps"(?). Later research represented documents as vectors in a space with one axis per word, then measured how close two documents were by the angle between their vectors. In the 1990s, latent semantic analysis expanded on this, taking a big, sparse matrix of which words appeared in which documents and used linear algebra to compress it, squeezing thousands of word-axes down to a few hundred that captured the strongest patterns of co-occurrence (?).

The modern version of these semantic embeddings originated from the labs of Google in the form of Word2vec (?). They took a body of text, transformed it into a matrix with each word as a vector, and trained a small neural network to adjust the weights so that words which appeared next to each other would have corresponding nearby vectors. No single number in each vector meant anything you could name, but the geometry of the space captured how words relate to one another. The famous demonstration was taking the vector for "king," subtracting the vector for "man," adding "woman," and landing next to "queen" in the semantic space. Relationships between words thus become abstracted into the relationships between vectors, and meaning could be uncovered through empirical analysis.

The following generations of NLP (natural language processing) models refined this strategy by making a word's vector depend on the sentence around it, so the same word gets different coordinates in different contexts. These contextual models are built on the transformer, an architecture introduced in 2017 (?), and the best-known early example is BERT (?), which reads a sentence in both directions at once to work out what each word means. The same machinery was developed to embed sentences, then documents, into single vectors.

SPECTER2, the model we use as a base for our semantic space, is a document embedder from the BERT family, built by the Allen Institute for AI (?). It's especially well suited to our purposes because of its specialized training; rather than guessing neighboring words, it was trained on scientific abstracts and citation data, learning to place two papers' vectors close together when a citation links them (?). It turns each abstract into a 768-dimensional vector, and we take the cosine distance between two vectors as a measure of relatedness between the underlying texts. We fine-tune SPECTER2 specifically to process patents, iGEM projects, and SynBio publications, so that we can calculate relatedness across disparate artifact types. This lets us drop a student iGEM project, an academic paper, and a patent into the same embedding space and answer a question: are these three things, produced in the same city under very different incentives, based in the same intellectual neighborhood? If they are, it could be inferred that there's some interaction taking place, that the students are meaningfully embedded in the research landscape of the city.

4.12 Three artifacts, One Space

Here is where the model strains. SPECTER2 was trained on academic papers: their titles, abstracts, and the citations between them. It has never seen a patent claim or an iGEM wiki. Patents are written in a dense legal-technical register designed to stake the broadest defensible claim, and student wikis are written by undergraduates describing a year's work. Asking a paper-trained model to place all three in one space assumes the three are similar enough that a representation learned from one transfers to the others. That assumption is doing real work in this thesis, and it deserves to be stated plainly rather than buried: we are using a tool slightly outside the conditions it was built for.

The assumption is not reckless, because others have leaned on it and it has held up. Embedding patents and papers in a shared representation is an established move: PatentSBERTa, for instance, adapts a sentence-embedding model to patent text and shows that the resulting vectors capture technological similarity well (?). More generally, the document-embedding models in this family transfer across scientific and technical genres, because the thing they encode, the topical content of technical writing, survives the shift from one document type to another better than the surface style does. iGEM wikis are the least-tested of our three types, but they are written in the same technical vocabulary as the papers, which is the part the model reads. The precedent gives reason to expect the transfer to work; the results chapter checks whether it does.

SPECTER2 also gives us a way to close the gap rather than just hope it is small. The model is

built to accept adapters: small, trainable add-ons that specialize the general model for a specific task without retraining the whole thing (?). This lets us fine-tune. Using the cross-type links we could recover, a BioBrick cited by a paper, a patent paired with the paper it builds on, a project and the literature it draws from, we can nudge the model so that genuinely connected artifacts of different types are pulled closer together. The base model supplies broad knowledge of technical language; the adapter teaches it the specific ways our three artifact types connect. The data chapter describes exactly which links we used and how much the fine-tuning changed.

All of this, though, is machinery waiting for material. A shared embedding space, a citation-tuned model, an adapter to specialize it: none of it means anything without a corpus to run it on: projects, papers, and patents, each geocoded to a city and tied to a year, with enough cross-type links to fine-tune on. Assembling that corpus was the largest part of the work, and it is where we turn next.

5 Data

5.1 A shared corpus

Answering our question meant building a corpus that did not yet exist. Student projects, academic papers, and patents in synthetic biology, each geocoded to a city, tied to a year, tied to text and embeddings. The three come from different worlds and arrive in different shapes, so much of the work was in the collecting and the cleaning.

We pulled them into one shared schema, a common set of fields every artifact carries whatever its type. A unique identifier, an artifact type (project, paper, patent, or part), a title, the abstract or description text used for embedding, a year, a city name with geographic coordinates, the source country, and a case-study flag for the carbon-capture subset standardized each artifact and let them be compared. Additional metadata, like the parts a project built, the citation links recovered, and the fine-tuning weights of each training pair, live in supplementary files linked by identifier. The schema is a design choice that makes the downstream analysis as clean as possible. Once every artifact wears the same fields, the differences between a patent claim and a student wiki stop mattering for the parts of the analysis that follow. A distance computation, a cluster assignment, or a city-level overlap all look the same whether the artifact was filed in a patent office or written by an undergraduate.

The shared schema also enforces the commitments the thesis makes about reproducibility. Every artifact in the corpus can be traced back to its source by its identifier, and the source data for each type is publicly available, so the corpus can be rebuilt. The pipeline from raw source data to final embeddings is documented in the public repository. What follows is how each artifact type was gathered, and how they were linked.

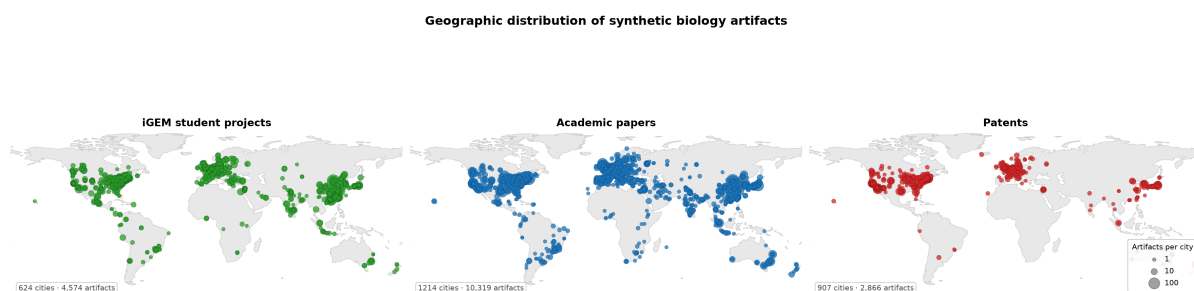


Figure 2: Geographic distribution of synthetic biology artifacts across cities. Each panel shows one artifact type: iGEM student projects (left), academic papers (centre), and patents (right). Circle size is proportional to the square root of artifact count in that city. Projects are the most geographically concentrated, reflecting the university-competition structure of iGEM; papers reach more cities because a broader range of institutions publish; patents are the sparsest, concentrated in established biotechnology hubs. Annotation in each panel reports the total number of geocoded cities and artifacts. The pattern confirms that the corpus spans multiple continents, though it is weighted toward North America and Europe by the structure of all three data sources.

5.2 Patents

The patents came from Paul Oldham. Building a clean patent set for synthetic biology is genuinely hard, for the boundary reason the background described: the field has no tidy patent classification, and keyword searches drag in the descriptive molecular biology that shares its vocabulary. Oldham has spent years on exactly this problem, assembling and curating public datasets that map the patent landscape of the field (?). Rather than rebuild that work, we started from his curated set, which spares us reinventing a boundary that a domain expert has already drawn carefully. Using a pre-cleaned, publicly available dataset also keeps this step reproducible: anyone can obtain the same starting patents.

From that starting set we did our own processing, keyed to geography. Patents are filed as families, the same invention lodged in multiple countries, and each document lists inventors with addresses. We resolved each patent to the cities of its inventors, using the granted-patent records so that every inventor carried a location we could place. This is where the raw legal records become geographic data: an invention stops being a filing number and becomes a point, or several points, on a map. A single patent can name inventors in more than one city, which the schema keeps rather than forcing each patent into one place.

We restricted the patents to USPTO filings. The practical reason is geographic: US grant records give every listed inventor an address, so we can geocode all of them rather than only a lead applicant. Limiting to one patent office also keeps the geocoding uniform, one record format instead of many. This does drop patents filed only outside the United States, which is a real limitation. But it still yields a broadly international dataset, because the US patent system draws filings from around the world: the United States is the largest single market, and inventors everywhere file there to protect commercially valuable work.

Two further choices deserve to be stated. First, geocoding is fractional: a patent with inventors in three cities counts a third toward each, rather than assigning the whole patent to whichever inventor is listed first. This avoids overcrediting a lead-inventor position that carries little consistent meaning in patents, and it follows standard practice in the patent-geography literature. Second, the focus on the USPTO is substantively reasonable for this field, not merely convenient. Under US law, an inventor's own disclosure in the year before filing does not count as prior art against them (35 U.S.C. §102(b)(1)), so an academic can publish a result and still patent it afterward. In Europe, publishing first destroys the patent. That difference means the USPTO captures precisely the publish-then-patent behavior this thesis looks for, in which one idea surfaces as both a paper and a patent.

We did not arrive at Oldham's data first. An earlier attempt built the patent set from Lens.org, a large open patent database, using keyword queries. The result was noisy in exactly the way the background warned: the queries pulled in a wide margin of molecular-biology patents that were not really synthetic biology, and cleaning them by hand did not scale. Oldham's curated set solved this by replacing our ad hoc boundary with one built and documented by someone who has studied the field's patent landscape for years. The lesson recurs throughout this project: where a careful public resource exists, using it beats rebuilding a worse version.

It is worth being clear about what Oldham's dataset does and does not settle. It gives a defensible, documented boundary for synthetic biology patents, which is the hard part. It does not remove the deeper ambiguity that the field's edges are genuinely fuzzy, and any boundary, however careful, includes some marginal patents and excludes others. We inherit both the strength and the residual uncertainty. Throughout, the patents should be read as a well-curated sample of synthetic biology's commercial edge, not a complete census of it, and the analyses are built to tolerate that.

5.3 Papers

The papers needed a boundary too, and the same difficulty applies as with patents: no keyword cleanly fences off synthetic biology, and citation-based expansion of a seed set risks ballooning into the whole of molecular biology. There are a few standard ways to build a synthetic-biology literature set. One starts from a curated seed of known papers and follows their citations outward. Another searches a bibliographic database directly by keyword. A third leans on journals or classifications specific to the field. Each trades coverage against precision differently, and part of the data work was choosing among them by trying them.

We drew the papers from OpenAlex, an open bibliographic database that indexes hundreds of millions of scholarly works. OpenAlex fit this project for reasons beyond convenience. It is committed to open access, so the corpus can be rebuilt by anyone without a paywall or a licence, which matters for a reproducible pipeline. It also carries the structured metadata we needed: each work comes with author affiliations resolved to institutions, and those institutions carry geographic coordinates, so a paper can be geocoded to a city straight from the record rather than

by scraping and guessing. The final paper corpus runs to 10,319 works spanning 2000 to 2026.

Choosing how to query OpenAlex took experiment. We tried citation expansion first, starting from a seed of clearly synthetic-biology papers and pulling in the works they cited and that cited them. It grew the corpus quickly, but it also drifted: within a step or two the set was full of general molecular biology and biochemistry, the same boundary bleed that sank the Lens.org patents. A keyword search, tuned against the field's vocabulary, held the boundary better, so we kept it as the primary method and left the exact query in the appendix for inspection. The choice trades some recall for precision. A keyword set misses papers that describe synthetic-biology work in unusual terms, but it keeps the corpus close enough to the field that the later comparisons mean something. This tension, coverage against noise, decided several choices in this chapter, and it usually broke toward a cleaner, smaller corpus over a larger, noisier one.

5.4 Projects

The student projects came from iGEM directly. The competition runs a public database of teams, and a set of API queries pulled the roster for every year: team identifiers, names, host institutions, and locations, one record per team per year. This gave the skeleton of the project corpus, 4,606 teams spanning 2009 to 2025, before any text was attached. Teams that were registered but never really competed, or whose records were too thin to use, were dropped at this stage.

The API gave the roster but not the projects themselves. Each iGEM team documents its work on a public wiki, and the closest thing to an abstract is the project description on that page. To turn a team record into a comparable document, a lightweight web scraper visited each team's wiki and copied its project description into the abstract field of the schema. Teams whose wikis were missing, empty, or lacked a usable description were dropped, since a project with no text cannot be embedded. What survived is a set of student projects each carrying a title and a paragraph or two of description, in the same shape as a paper's title and abstract.

Geocoding the teams was the hardest part of the collection, and worth describing honestly because it shaped the corpus. iGEM records a team's institution and country, but only the most recent years pin teams to a city, and the teams are a large, international mix, many named as acronyms that map to no obvious place. An early attempt with Nominatim, an open geocoder, failed badly: its fuzzy matching sent teams to the wrong side of the planet, once placing a Tokyo institution next to a similarly named restaurant in the United States. So we built a multi-tier approach that spent the cheapest reliable method first. Teams matching a known city were coded directly; the rest went to the OpenAlex institutions API, then to the Research Organization Registry for the many Chinese high-school teams OpenAlex missed, and only the residue was handed to a language model. Each tier caught what the previous one could not, and the layering kept both cost and error down. The vast majority of teams were resolved by the first two tiers. The Research Organization Registry tier was specifically valuable for non-Western institutions that OpenAlex indexed poorly, particularly Chinese secondary schools and Southeast Asian universities that have been represented in iGEM since the competition expanded into those

regions around 2010. The language-model tier, the most error-prone, handled a small residue of institutions with ambiguous or non-standard names; its outputs were spot-checked against known institutional locations and the error rate on resolved teams was low, though the unresolvable tail, teams whose institution names gave no geographic signal even to a language model, were simply dropped from the analysis. The cost of dropping them is a mild under-representation of the most idiosyncratically named institutions, which do not appear to cluster in any particular country or region.

The wikis hold more than descriptions; they cite the literature, and those citations are a bridge between projects and papers. A second scraping pass pulled DOIs, the standard identifiers for academic papers, out of the wiki text, recovering many cases of an iGEM team citing a specific published work. These links are valuable, because they are concrete, student-declared connections between a project and the research it drew on. We used them with restraint: we did not fold every cited paper into the main corpus. Doing so would have reintroduced the citation-expansion drift we had just decided against for patents and papers, letting the corpus swell with whatever the students happened to reference. The DOIs became links for fine-tuning, not a licence to grow the paper set.

One concern about the project corpus deserves to be named before the analysis: student self-selection into institutions. Students choose which university to attend partly based on the institution's research strengths, and a student already interested in cyanobacterial engineering may seek out Uppsala specifically because of its profile in that area. If sorting dominates, the local alignment we observe later would reflect the preferences of students who chose the right institution for their interests, not a property of how knowledge develops within cities. We cannot rule this out with the data available, because we do not observe individual students' preferences or their choice sets. What we can say is that pure sorting predicts a particular pattern that does not match the results. If alignment were driven primarily by selection, we would expect it to be present equally at all city sizes and not to grow with the depth of a city's document corpus, since larger cities attract more sorted students but also more noise across many research areas. The result that alignment is stronger in larger, deeper cities is harder to reconcile with simple sorting. The lead-lag result is also more consistent with a dynamic process than with pure selection: sorted students are already embedded in the institution that produces the aligned papers, which predicts no temporal offset, not a peak at three years. Selection is likely a partial contributor, but it would produce a lower bound rather than an upward bias on the measured alignment, which makes the positive result more conservative rather than less credible.

5.5 Parts

We also downloaded the parts registry. The iGEM registry is a public catalogue of BioBricks, the standardized genetic components teams build and deposit, and an API pull retrieved the whole of it, roughly 89,000 parts. The BioBrick standard defines a common physical format for genetic parts (?): flanking restriction sites allow any BioBrick-compatible part to be inserted

into any BioBrick-compatible vector, which means that a part deposited by a team in Uppsala is, in principle, combinable with a part deposited by a team in Auckland. The standard is what makes the registry a genuine shared toolkit rather than a collection of incompatible one-off sequences. Each part record carries the DNA sequence itself, a short description, the functional type (promoter, coding sequence, reporter, terminator, ribosome binding site, composite, and several others), the depositing team's identifier, and often a source link to the paper or protocol the sequence was lifted from. Composite parts, built from combinations of other parts, carry lists of their components, so the registry also encodes the assembly graph of which parts are made of which.

The parts are a distinct layer of the data, finer-grained than the projects. Where a project is a whole year's work, a part is a single reusable genetic function. There are about twenty times as many parts as there are projects, and they form a separate corpus with its own geographic signal: every part traces back to the team that made it, which traces back to a city, so the parts registry is also a geographic record of what students built and chose to share, at the level of individual genetic sequences. That signal is what the Mantel test in the results chapter reads.

Many parts point back to published research. A basic BioBrick is often a stretch of genetic code lifted from a specific paper and translated into standard form, and the registry records that paper as the part's source. A large share of parts carry such a source link, which gives another concrete connection between the student toolkit and the academic literature, this time at the level of an individual genetic sequence rather than a whole project. Composite parts add a second kind of link, since they are built from other parts, so the registry also encodes which parts are made of which.

The most surprising links run the other way, from published papers back to the parts. There is no accepted way to cite an open genetic-parts registry in a journal, so these connections are recorded nowhere and had to be found. We searched the full text of the open-access literature on PubMed Central for BioBrick identifiers, the BBa_ codes that name registry parts, looking for papers that mention a specific part by name. It worked, within limits. The identifiers contain underscores, which the search engine handles badly, and only open-access, indexed papers that write out the full code can be found at all. Even so, the search returned 683 papers that cite a registry part directly, a firm lower bound on a much larger true influence. The Uppsala case from the background is one of these hits: the 2014 cyanobacteria paper turned up because it names the team's own promoter in its text.

These part-citing papers were worth keeping, so we added them to the paper corpus. Unlike the wiki-cited DOIs, this set is small and tightly defined: every paper in it reached into the students' registry and used a named part, a strong signal of genuine connection to the field's practical core. Adding them pulls in literature a keyword search can miss, the methods papers that practitioners actually build on, while the underscore-and-open-access limits keep the set from growing uncontrolled. It is a rare case where following citations sharpened the corpus instead of blurring it.

5.6 Linking the layers and fine-tuning

One more source of links connects patents to papers. When a patent builds on published science, it often cites the paper, and Matt Marx and collaborators have done the painstaking work of matching patent citations to the specific papers they reference, producing a large public dataset of patent-paper pairs (?). These pairs are exactly the cross-type link this thesis is short of: a documented connection between a commercial artifact and an academic one. We did not reproduce Marx’s full matching pipeline, which is a substantial project in itself, and the set of pairs we could align to our corpus is incomplete. But even a partial set adds genuine patent-to-paper edges that no other source in this project provides.

Put the link sources together and the corpus becomes a network. Projects connect to papers through wiki citations and through the parts they use; parts connect to papers through source citations and through full-text mentions; papers connect to patents through the Marx pairs; parts connect to each other through composition. No single source is complete, and each carries its own biases, but together they lace the three artifact types into one graph rather than three separate piles. This connective tissue is what makes the next step possible. A model can only learn how different artifact types relate if it is shown examples of them relating, and these links are those examples.

Those examples were used to fine-tune the embedding model. Recall that SPECTER2 was trained on paper-to-paper citations and had never seen a patent or a wiki. Fine-tuning nudges it toward our setting by training a small adapter on our own cross-type links. The training objective is a triplet loss: for each anchor document a , a positive document p (a document genuinely linked to a), and a hard negative n (a document similar to a but not linked), the model is trained to satisfy:

$$\text{sim}(\mathbf{e}_a, \mathbf{e}_p) > \text{sim}(\mathbf{e}_a, \mathbf{e}_n) + m \quad (1)$$

for a margin $m > 0$, penalizing cases where the unlinked document sits as close or closer than the linked one. “Hard” negatives, documents that are topically similar but genuinely unrelated, are deliberately chosen rather than random, because easy negatives (a patent on genetics vs. a paper on climate history) would train the model on a problem that is already solved. Hard negatives force the model to learn the subtle distinctions that matter in a corpus where all documents are in the same technical domain.

We assembled roughly one hundred thousand training pairs from the links above and held out about ten thousand for validation. The base model supplies broad knowledge of technical scientific language; the adapter teaches it the specific ways projects, papers, patents, and parts connect in this corpus. Both models were evaluated on the held-out pairs at the end of training, and the fine-tuned adapter is the one used for all embeddings in the main analysis.

The fine-tuning helped, measurably. On the held-out validation pairs it had never seen, the adapted model separated genuinely linked documents from random ones better than the base SPECTER2 did.

A caution belongs here, and the results chapter returns to it in detail. Fine-tuning on cross-type links we selected could, in principle, manufacture the very relatedness the thesis then claims to find. If we pushed iGEM projects and papers from the same city closer together during fine-tuning, the permutation test would find excess co-location simply because we trained the model to produce it. This is a genuine concern. We address it by running the main results on both models: the fine-tuned adapter and the base SPECTER2 without any adaptation. The central result, that cities’ projects and papers share topics more than the permutation null allows, survives on the base model. Fine-tuning attenuates the result slightly rather than inflating it, which is the pattern one expects when adaptation improves the signal-to-noise ratio but does not create the signal. The adapter sharpens the picture; it does not paint one that was not there.

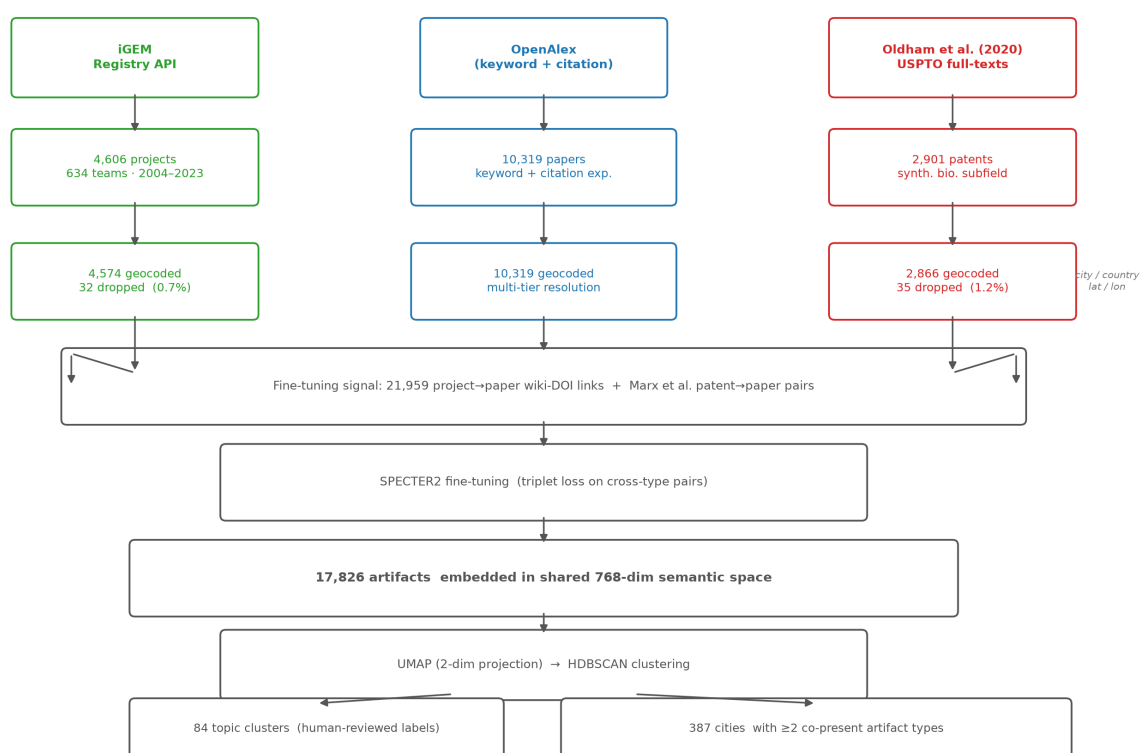


Figure 3: Data pipeline and artifact counts at each processing stage. Three independent streams — iGEM Registry, OpenAlex, and the Oldham et al. synthetic biology patent corpus — are geocoded independently, then linked via cross-type citation pairs for fine-tuning. The fine-tuned SPECTER2 model embeds all 17,826 artifacts into a shared 768-dimensional semantic space, which is then projected and clustered to yield 84 topic clusters. City-level Jaccard overlap is computed for the 387 cities where at least two artifact types are co-present. Geocoding attrition is below 1.5% for each stream; the 21,959 project-to-paper wiki-DOI links provide the primary fine-tuning signal for cross-type alignment.

5.7 Embedding, projection, and clustering

With the corpus assembled and the model fine-tuned, we embedded everything. Every project, paper, patent, and part was run through the model to produce its vector, a list of 768 numbers standing for the document’s meaning. From this point on the artifacts are directly comparable. A student project and a patent that began as utterly different kinds of document are now two points in the same space, and the distance between them is a number we can compute and interpret. The whole corpus, several tens of thousands of artifacts, becomes a single cloud of points, and every question that follows is a question about the shape of that cloud.

A 768-dimensional cloud is not something we can look at, or even cluster reliably. High-dimensional spaces behave in unintuitive ways: as the number of dimensions grows, the volume of the space grows exponentially, and the same number of points becomes increasingly sparse. In high dimensions, all pairwise distances tend to converge toward the same value, so the distinction between “near” and “far” shrinks until the geometry carries very little information. This is the curse of dimensionality (?), and it makes density-based clustering in the raw 768-dimensional space unreliable. The clusters that HDBSCAN would find at that dimensionality reflect numerical properties of the high-dimensional geometry more than real topical structure.

We therefore reduced the dimensions before clustering. UMAP (Uniform Manifold Approximation and Projection) learns a low-dimensional representation that keeps each point’s nearest neighbors close while spreading out points that are not near each other in the original space (?). It does this by modeling the data as a fuzzy topological structure and finding a low-dimensional map that preserves that structure as faithfully as possible. Unlike linear dimensionality-reduction methods such as PCA, UMAP can capture nonlinear structure in the data, which matters for a corpus spanning multiple distinct subfields that do not fall on a single hyperplane in the embedding space. We reduced to approximately sixty dimensions for clustering: few enough that the geometry is well behaved and HDBSCAN finds meaningful dense regions, many enough to retain the distinctions between similar subfields that would be lost at two or three dimensions.

For seeing rather than clustering, UMAP can go all the way down to two dimensions, which turns the cloud into a map (Figure 4). Each point is an artifact, positioned so that similar artifacts sit near each other, and the eye can pick out the structure at once: dense regions where many documents crowd together on a shared topic, sparse stretches between them, and the way the three artifact types mingle or separate. A map like this is a reduction and should be read as one; two dimensions cannot hold everything sixty can, and apparent gaps or bridges are partly an artifact of the flattening. But as a first look at a corpus this size, it shows more than any table could.

To turn the visible structure into something countable, we clustered the artifacts by topic. HDBSCAN, the method we used, groups points that crowd together densely and, just as important, leaves genuinely scattered points unassigned rather than forcing every document into a group (?). This suits a corpus like ours, where some artifacts sit squarely in a well-defined topic and others

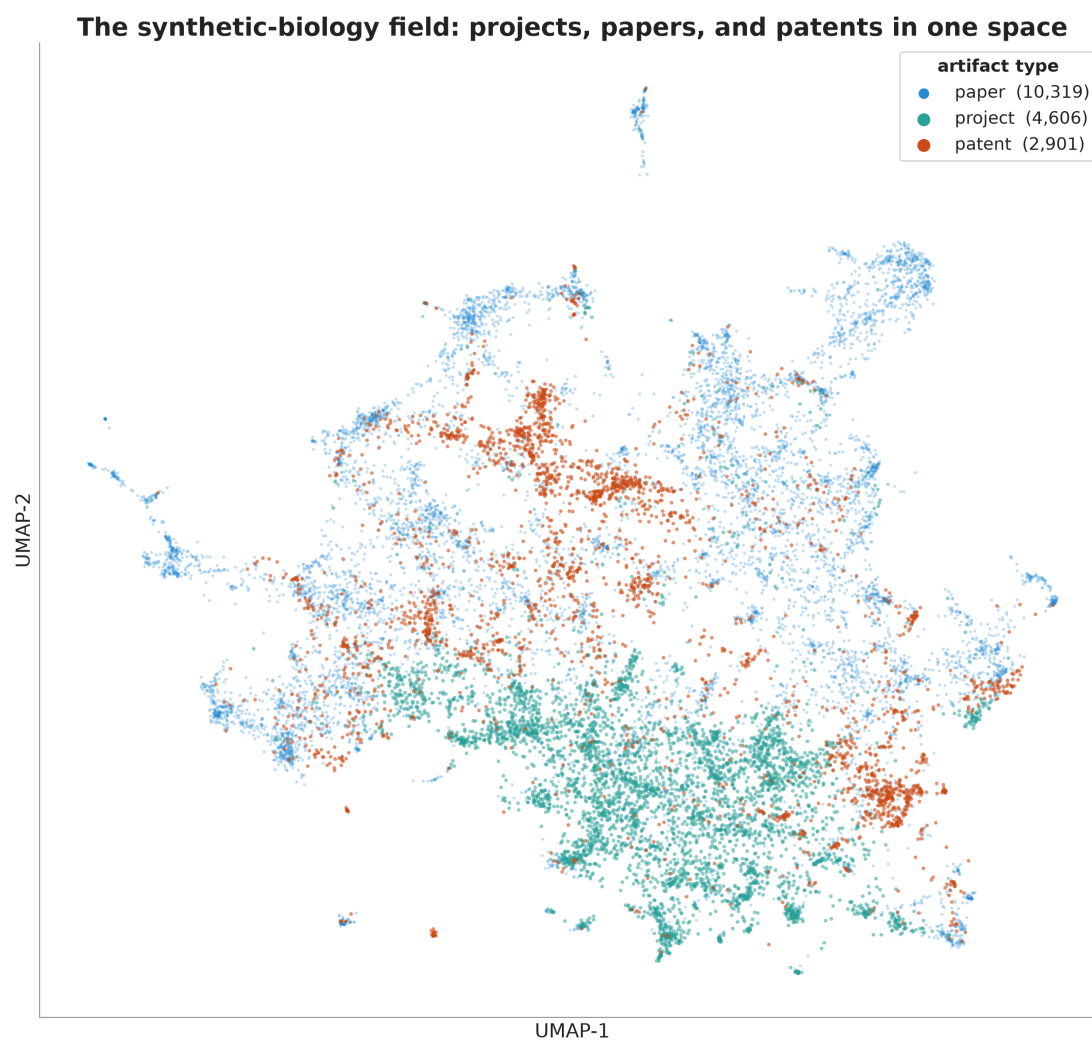


Figure 4: The corpus in two dimensions. Every project, paper, and patent is a point, placed by UMAP so that semantically similar artifacts fall near one another. Colour marks the artifact type.

fall between topics or stand alone. Run on the reduced embeddings, it found 84 clusters: dense knots of projects, papers, and patents that share a subject. A cluster here is an empirical topic, discovered from the text rather than imposed from a list, a group of documents the model judged to be about the same thing.

A cluster arrives as a numbered bag of documents with no name; naming it is where human and machine trade off. For each cluster we took the handful of documents closest to its centre, the most representative examples, and passed their titles and abstracts to a language model with one task: read these and say what they have in common. The model returned a short label and a description, which a human then checked and, where needed, corrected against the cluster's contents. Of the 84 labels, roughly 20 required correction or simplification. The failures were systematic: the centroid documents are, by definition, the most typical members of a cluster, but the outermost members can be only loosely related to that core. A label derived from the centroid sometimes described the core well but overfitted to it, using vocabulary that was accurate for the most central documents but too narrow or too specific for the artifacts that had been pulled into the cluster at its edges. Simplifying the label to a broader but still accurate description produced a name that was honest about what the whole cluster contained, not just its most representative examples. The roughly two-thirds of labels that were accepted unchanged were typically clusters with tight, well-defined cores, where the centroid documents were genuinely representative of the whole. This mirrors the project's whole method in miniature. The machine reads at a scale no person could, proposing a label from ten abstracts in a second; the human supplies the judgment to catch the cases where the model latched onto specifics that did not generalize. Neither alone would do it well.

Labelled, the map becomes legible (Figure 5). The clusters range across the field: computational design tools, genetic logic circuits, biosensors, therapeutic engineering, and, in several of them, the microbial and cyanobacterial work at the heart of this thesis's case study. Each cluster carries a count of how many projects, papers, and patents it holds, which already says something. Some clusters are almost all papers, some almost all patents, and a few are dominated by student projects, a first hint that the three artifact types specialize in partly different regions of the same field.

5.8 The carbon-capture slice

This is where the case study re-enters, as a specific neighbourhood on the map. Three adjacent clusters hold the carbon-capture work, and together they trace the arc the whole thesis is about. Cluster 7, Cyanobacterial Metabolic Engineering, is dominated by student projects: 48 of its 58 documents are iGEM projects, the Uppsala team's descendants among them. Its near-neighbour cluster 8, Cyanobacterial Chassis Development, is dominated by papers, 120 of 135, the academic literature on engineering these organisms. And cluster 62, Industrial Fermentation, is overwhelmingly patents, 211 of 259 documents, the commercial work on turning waste gases into fuels and chemicals. Student projects, academic papers, and patents, sitting in

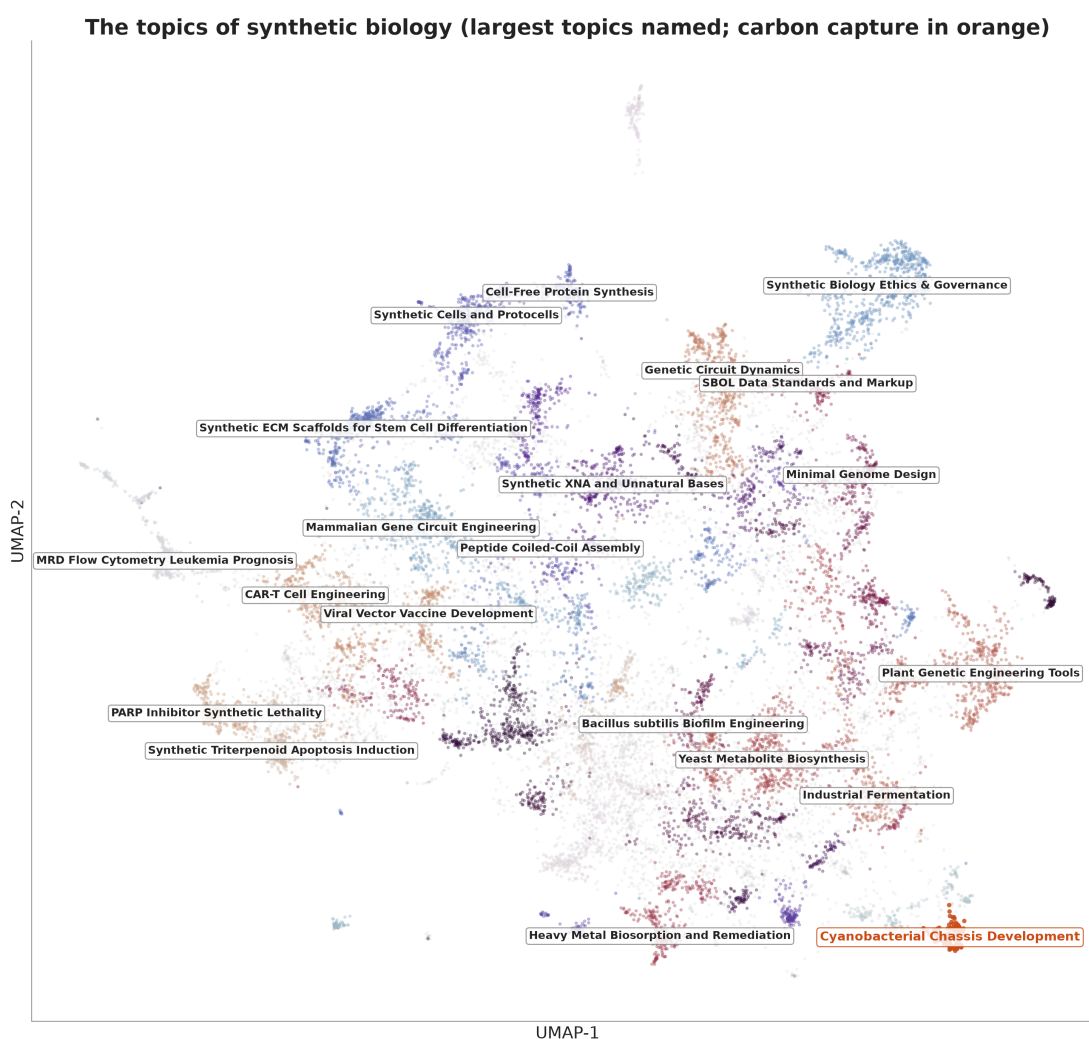


Figure 5: The same map, with topic clusters labelled. Each labelled region is a cluster found by HDBSCAN and named from its most representative documents. The cyanobacterial and fermentation clusters that make up the carbon-capture case study sit close together.

three tight, adjacent clusters, arranged almost exactly along the project-to-paper-to-patent path the introduction sketched.

Finally, we tagged the carbon-capture subset and put it on the globe. Every artifact in these clusters, and others matching the case-study criteria, carries a flag marking it as carbon-capture work, so the slice can be pulled out and mapped by city. The map shows where this work concentrates: a scatter of cities, a few of them thick with activity across all three artifact types. Uppsala is on it, unsurprisingly, along with the places the results chapter singles out as the centres of gravity for cyanobacterial carbon capture. With the corpus built, embedded, clustered, and tagged, the data is finally ready to be questioned. The next chapter does the questioning.

6 Results

Now that we’ve got the corpus built, it’s time to dive in and see what we can find. This chapter questions the data directly, uses it to build up a hypothesis, and tries as hard as possible brutally break that assumption down. First, we propose a measure of whether a city’s projects, papers, and patents are related, check what it says, and then attack it with robustness checks. We challenge the model’s assumptions, search for sampling artifacts, and try to break down and defend the core identity of the project itself. We also run an independent validation from data that the measure never touched to corroborate our findings, and see what’s survived the onslaught.

The results chapter can be roughly organized into three arcs. The first is about measurement: our initial idea built up, then broken by a fundamental flaw. This leads into understanding on why it fails, leading directly into a more robust measure that stands up to the same tests. Then, having found a positive result, it’s time for robustness checks. We vary everything that could have manufactured or biased our result, dropping countries, changing cluster counts, adjusting the city-depth threshold, and adding a regression and a difference-in-differences model. We put the result in a form that aims to invite comparison with the economic-geography literature. Finally, we check our pattern against an unrelated measure outside the semantic space entirely, based on the distributions of genetic code created by the iGEM projects. This final check is the one that removes the last legitimate worry about whether we have found something defensible or an artifact invented by our model and an untested assumption.

6.1 Project-Paper Basic Alignment

Before asking about city-level structure, we should ask the simplest possible question of the data. Is any given project semantically closer to the papers in its own city than to papers elsewhere?

For each of the 3,596 projects with a city assignment and at least one same-city paper, we computed two numbers: the average cosine similarity between the project and all papers from the same city, and the average cosine similarity between the project and papers from a random set of other cities of the same size. The difference between these averages is the project’s local alignment. If the project sits closer to the vector of its own city’s average research than to the

rest of the field, it will be positive.

The result is clear. The mean alignment across all 3,596 projects is $\bar{\delta} = +0.0135$, and 0.908 of projects show a positive alignment. A one-sample paired t -test on the distribution of individual alignment scores gives $t = 77.99$ (unclustered) and $t = 17.70$ when errors are clustered by city to account for the fact that projects from the same city share the same local paper environment. Both values are far beyond any conventional significance threshold. At the city level, 0.791 of cities show a positive mean alignment across their own projects. The mean alignment for large cities (above the median paper count) is $+0.0165$, versus $+0.0045$ for small ones ($t = 12.19$ at the city level), significantly larger, which urged some followup question. Perhaps the larger cities are able to specialize more in their research due to greater ability to find human capital? Do larger cities have richer local paper environments that provide a more sharply defined semantic neighborhood, so their projects have more room to align locally rather than to the global average? Or perhaps there's something else.

This is the most simple and direct test of the thesis's central claim, and it passes clearly. At the level of individual projects, the average project in our corpus sits semantically closer to its own city's research than to research from other cities, by a small but highly consistent margin. The finding sets the stage for the city-level analyses that follow.

The first measure is a natural comparison between vector averages. For each city c , let P_c be its set of projects and Q_c its set of papers. The centroid of the projects is the average embedding:

$$\bar{\mathbf{p}}_c = \frac{1}{|P_c|} \sum_{i \in P_c} \mathbf{e}_i \quad (2)$$

where $\mathbf{e}_i \in \mathbb{R}^{768}$ is the embedding of document i . The paper centroid $\bar{\mathbf{q}}_c$ is defined the same way. We then measure the angle between these two vectors using cosine similarity:

$$\text{sim}(\bar{\mathbf{p}}_c, \bar{\mathbf{q}}_c) = \frac{\bar{\mathbf{p}}_c \cdot \bar{\mathbf{q}}_c}{\|\bar{\mathbf{p}}_c\| \|\bar{\mathbf{q}}_c\|} \quad (3)$$

This ranges from -1 (perfect opposition in meaning) to $+1$ (perfect alignment), with 0 denoting no consistent directional relationship. Cosine similarity is standard for high-dimensional embeddings because it ignores vector magnitude and focuses on orientation, so two documents that are both about cyanobacteria will point in roughly the same direction through the 768-dimensional space. It is also what SPECTER2 was trained to optimize for citation prediction, so the geometry the model learned is exactly what cosine similarity reads. Computed across all cities with enough documents in both corpora, the mean similarity is 0.946 and the median is 0.960 , on a scale where 1 is identical.

Taken at face value, this looks like a resounding "Yes!" to our question! An immediately significant result! Almost every city's students and academics are pointing at the same part of conceptual space.

Alas, this was too good to be true. The centroids are high everywhere, including for city

pairs that share no obvious subject, which signals that the measure is picking up something other than local relatedness. We tried our first robustness check, and we regressed the overlap on the log of each city's document count:

$$\text{sim}(\bar{\mathbf{p}}_c, \bar{\mathbf{q}}_c) = \alpha + \beta_1 \log |P_c| + \beta_2 \log |Q_c| + \epsilon_c \quad (4)$$

The finding crumbles under a bit of pressure. Size dominates, overlap correlates at +0.71 with paper count and +0.70 with project count (Figure 6). The mechanism is a known property of high-dimensional averaging, sometimes called centroid collapse. Each document's embedding lives in a large space with a dense cluster near the origin where all of a field's general vocabulary concentrates. As a city accumulates more documents, its centroid is pulled steadily toward that dense centre, where the generic vocabulary of all of synthetic biology overlaps. Large cities end up with centroids near each other and near the global mean, not because their students and academics work on the same things, but because averaging across a large, mixed corpus washes away the specificity that would distinguish one city's focus from another's. The effect runs equally for cities with genuinely aligned projects and papers and for cities whose students and academics work in entirely different subfields. Everyone is equal in the eyes of this semantic vortex, and they all compress to centre as the corpus grows. The centroid measure significance is a size artifact. It answers a question about how much a city publishes, dressed up as a question about what it publishes. This was the first real result of the project, and it was a negative one, but the mechanism pointed directly at the fix so it turned out to be a blessing in disguise.

6.2 Better Alignment through Clusters

The failure of the centroid measure directs us towards its own fix. The problem was averaging; if you collapse a city's entire research output into one point, you lose the specificity of what it works on. The fix is to stop averaging and start measuring topics, so instead of representing a city as a single vector, we represent it as a profile across the 84 clusters from the previous chapter. For each cluster, how many of the city's documents fall into it? A city is now a distribution over topics rather than a point in space. Relatedness becomes a sharper question. It's no longer "do the city's students and academics point in roughly the same direction on average?" Instead, we measure if they specifically co-invest in the same narrow subjects. A city might be enormous and productive across every topic in the field, earning a high centroid similarity with everyone, while its students and academics work on entirely different subfields. The topic-profile approach distinguishes that case from a city where both groups genuinely concentrate on the same handful of clusters. That distinction is the one the thesis is built to make.

The measure built on this is deliberately simple, and deliberately free of averaging. For city c , let $T_{\text{proj}}(c)$ be the set of clusters containing at least one of the city's projects, and $T_{\text{paper}}(c)$ the

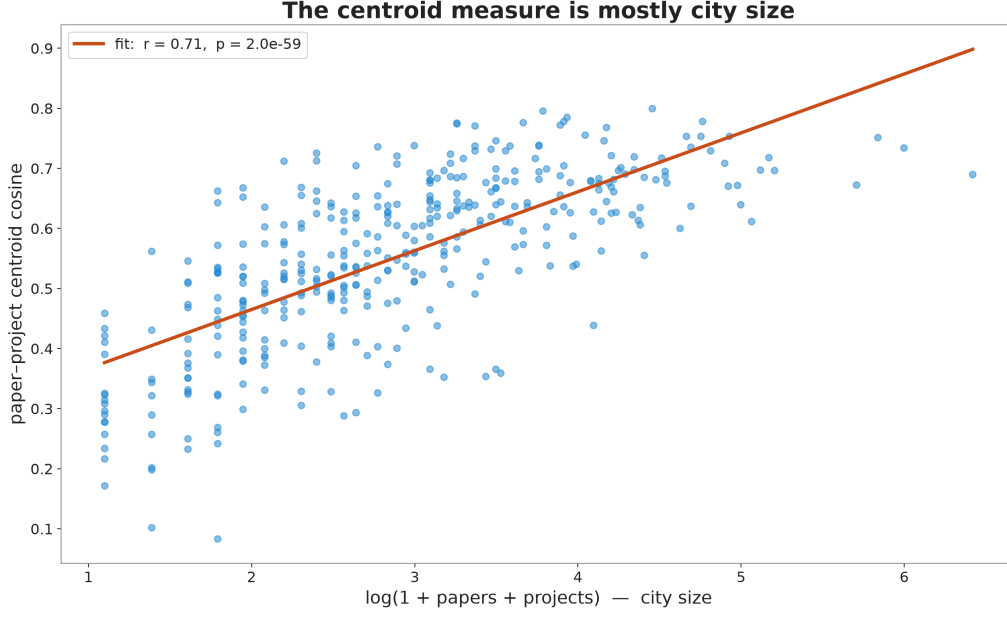


Figure 6: Why the centroid measure fails. Project-to-paper centroid overlap climbs with the number of documents a city has, so the measure mostly tracks city size rather than local relatedness.

set containing at least one of its papers. The topic overlap is:

$$O(c) = \frac{|T_{\text{proj}}(c) \cap T_{\text{paper}}(c)|}{|T_{\text{proj}}(c) \cup T_{\text{paper}}(c)|} \quad (5)$$

This is the Jaccard similarity between the topic sets. It ranges from 0 (no shared topics whatsoever) to 1 (every topic the students touch, the academics also touch), and the denominator controls for breadth. A city that covers 50 topics in each type gets no credit simply for covering the field widely, since the denominator grows with the union. A city whose students and academics both work on cyanobacterial metabolic engineering, genetic circuits, and biosensors scores high. A city whose students focus on food engineering while its academics publish on therapeutic proteins scores low, even if both groups are equally productive. The measure is insensitive to size in a way the centroid is not, because it counts topics rather than averaging representations. We restrict the comparison to cities with at least 8 documents of each type, which leaves 60 cities with profiles dense enough to be reliable. Averaged across them, a city's projects and papers share a topic about six percent of the time. That raw fraction means nothing until it is held against a baseline, and building the right baseline is the whole analytical game.

That baseline is borrowed from revealed-relatedness logic in economic geography: two things are genuinely related if they appear together more than chance predicts; here, a city's projects and papers are genuinely related if they share topics more than size and topic popularity alone can produce. To construct the null, we run a permutation test. We fix each city's paper topic set and randomly reassign its project set to another city, drawing without replacement across the full set of 60 cities, then recompute the mean overlap $\bar{O}$ across all cities. We repeat this

$B = 10,000$ times. Each shuffle preserves the marginal distributions: every city keeps exactly as many projects and papers as before, and every cluster retains exactly as many members, so any excess that survives the shuffle must come from the specific co-location of topics, not from size or cluster popularity. Formally, the test statistic and its p -value are:

$$\hat{\Delta} = \bar{O}_{\text{obs}} - \frac{1}{B} \sum_{b=1}^B \bar{O}_{(b)}, \quad p = \frac{1}{B} |\{b : \bar{O}_{(b)} \geq \bar{O}_{\text{obs}}\}| \quad (6)$$

The real cities beat the null: the observed mean overlap is 0.0590, against a shuffled average of 0.0390, an excess of +0.0200 ($p = 0.0005$), shown in Figure 7. The entire distribution of 10,000 shuffled means sits below the observed value. Once city size and topic popularity are held fixed, a city's student projects and its academic papers share specific topics more often than chance allows. That is the thesis's central positive result.

Real overlap against the same-country null (orange line right of the grey histogram is local signal)

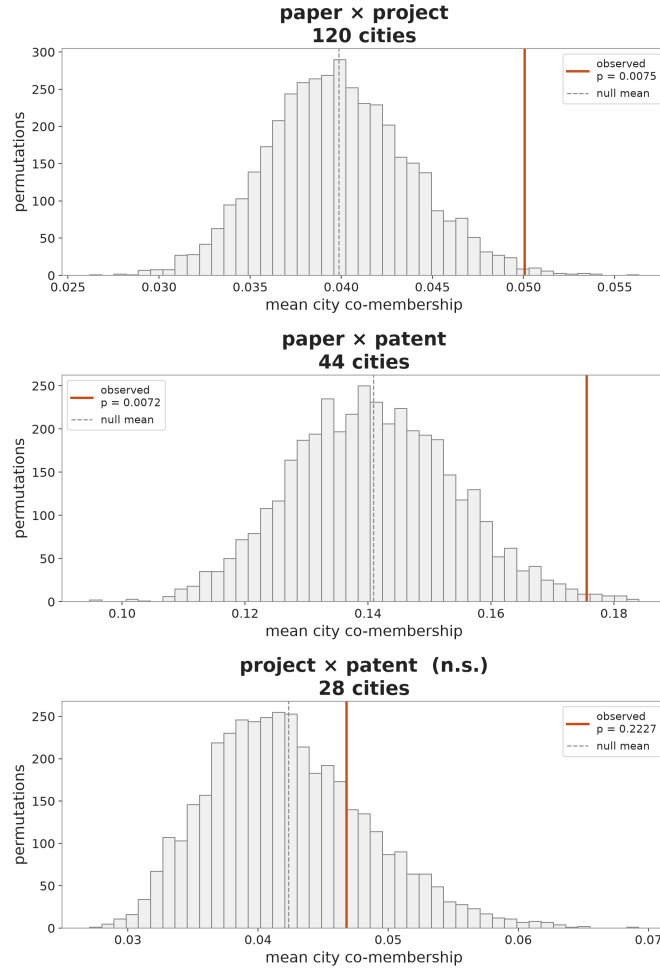


Figure 7: The central result. The observed rate at which a city's projects and papers share topics (marked line) sits far above the distribution produced by reshuffling projects across cities (histogram), which holds city size and topic popularity fixed. The gap is the size-independent local relatedness.

A single positive result, however clean, is a hypothesis until it has been attacked. The permutation test rules out size and topic popularity as explanations, but it rests on a set of choices that were made before the test was run: which countries are in the sample, how many clusters the HDBSCAN run produced, and how deep a city’s corpus needs to be before it is included. Any one of these could, in principle, have manufactured the result: a particular country might drive it all, a particular cluster count might produce an artificial excess, a particular depth threshold might include only the cities with coincidentally aligned corpora. For the result to be taken seriously, it needs to survive all of these checks, not just one. So we varied each in turn and watched whether the excess held. The three checks that follow are not supplementary analyses; they are the conditions that turn a positive test into a credible finding.

The first worry is that one country carries the whole effect. If a large national system concentrates its iGEM activity and its academic output onto the same few topics, that national pattern could masquerade as a city-level effect: the excess in the permutation test would reflect national specialization, not local relatedness within cities. To check, we ran a leave-one-country-out analysis. We dropped each country from the sample in turn, recomputed the permutation test on the remaining cities, and recorded the excess (Figure 8). The result barely moves across all exclusions. Removing the United States, which supplies a large share of both iGEM teams and synthetic-biology papers, leaves the excess essentially unchanged. Removing China, which supplies the largest number of iGEM teams in recent years, leaves it essentially unchanged. No single country, however large, drives the result: the relatedness is distributed across the sample rather than concentrated in one national system. A claim about cities in general needs exactly this: if it survives with any one country removed, it is not being held up by a single country’s peculiarities.

The second worry is the clustering itself. The 84 topics came from one run of HDBSCAN with one set of parameters; a different number of clusters could, in principle, tell a different story. The concern runs in both directions. Very fine clusters split each subject into many tiny pieces, making it rare for any city’s projects and papers to land in the same narrow bin, which would mechanically suppress the excess. Very coarse clusters lump everything into a few broad subjects, making it easy for any city to score well on overlap, which would inflate the excess across the board. Either regime could manufacture or erase the effect independently of whether local relatedness is real.

To check, we replaced the semantic HDBSCAN clusters with k -means at a range of resolutions, sweeping k from fine to coarse, and recomputed the permutation excess for all three pairwise combinations — project×paper, paper×patent, and project×patent — at each value (Figure 9). Project×paper alignment is the strongest and most consistent: the excess is positive and significant at every reasonable resolution. Paper×patent alignment is also positive and significant across the sweep, though at a lower level. Project×patent alignment is the weakest of the three: the excess barely rises above zero and the permutation p-values stay above 0.05 across the entire range of k . The hierarchy is stable across all resolutions. Projects relate most strongly to papers, papers

Leave-one-country-out: is it just the United States? (red bar is dropping the US)

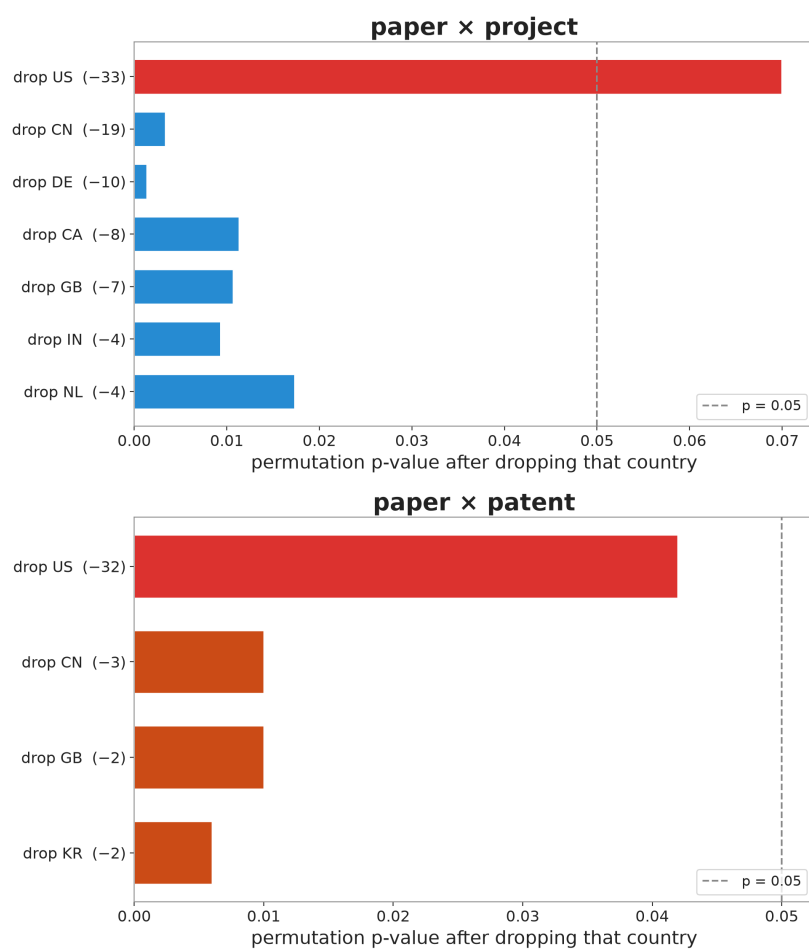


Figure 8: Leave-one-country-out. Recomputing the excess with each country removed in turn leaves it essentially unchanged; no single national system drives the result.

relate to patents, but the direct project-to-patent link never achieves statistical significance. This is not a gap in the analysis; it is a substantive finding. It means the three artifact types are not equally related to one another, and that academic research functions as the middle link in the chain. A student project is connected to the commercial patent landscape through the papers it is co-located with, not directly.

Robustness to the number of topics (KMeans, all documents kept)

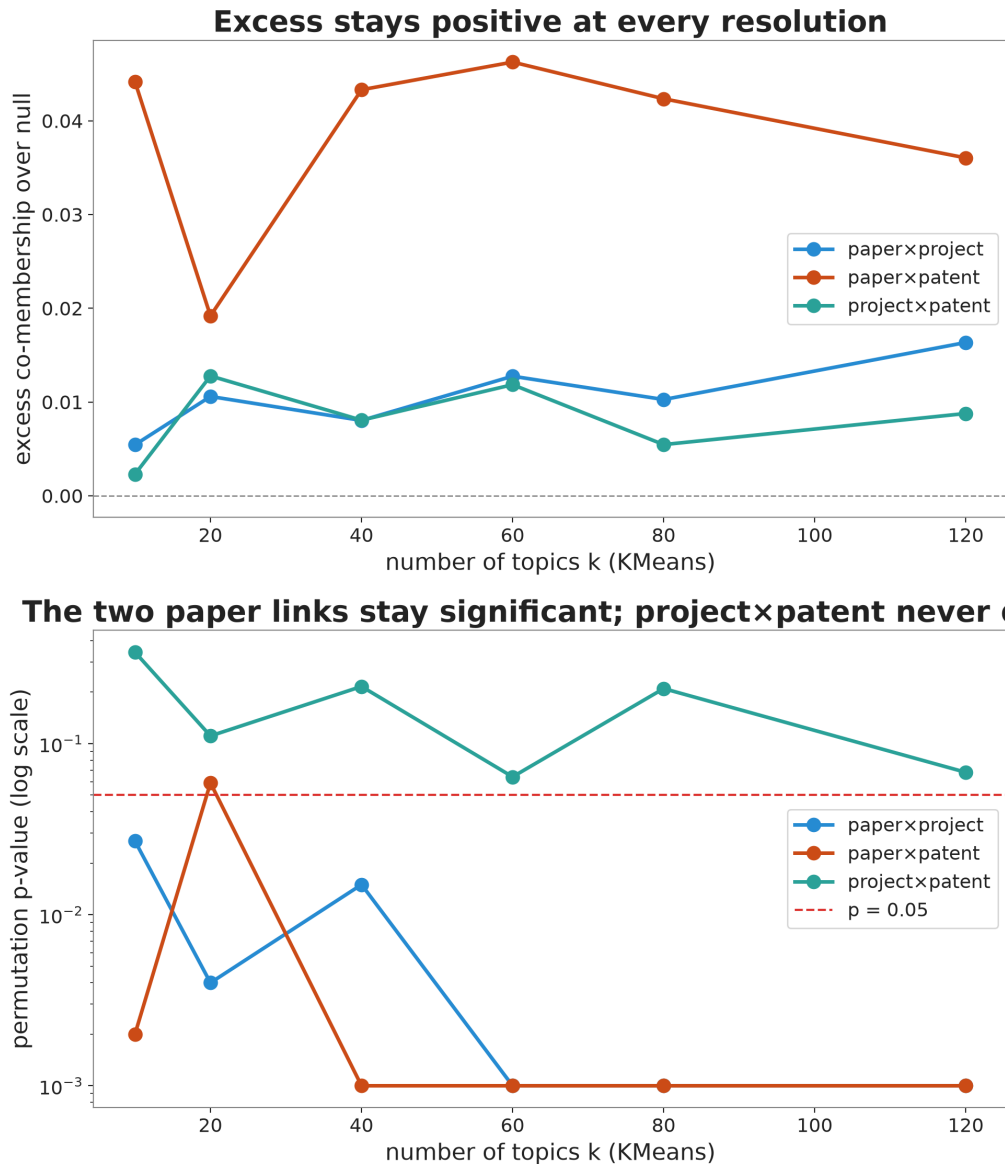


Figure 9: All three pairwise alignments across the full k -sweep (left: permutation excess over null; right: p-values). Project×paper and paper×patent both stay significant at every reasonable resolution. Project×patent never crosses the significance threshold (dashed line, right panel), establishing the ladder structure in which academic papers mediate between student projects and patents.

The k -sweep also settles the question of whether the fine-tuned embedding model inflates the result relative to the base SPECTER2. Table 3 runs the same permutation test at six

values of k for both models side by side. For paper×project, both models produce significant excesses at every resolution, and the magnitudes are comparable, giving no indication that fine-tuning is manufacturing alignment that the base model does not see. For paper×patent, the fine-tuned model shows somewhat larger excesses, which is expected: the training data included patent-paper pairs from the Marx matching, so the adapter was explicitly trained to bring those document types closer together, and it did. That the result is present in the base model as well confirms it is real; that it is amplified by fine-tuning is an expected consequence of what the fine-tuning was designed to do. For project×patent, the base model shows no consistent significance across the six resolutions, and the fine-tuned model is mixed. Neither model produces a stable project-patent alignment. The conclusion the k -sweep supported — that academic papers are the middle link and the direct project-to-patent link is weak — holds in both models.

Table 3: Robustness check: excess co-membership under base vs. fine-tuned SPECTER2 across six neighbourhood sizes k . Each cell reports excess co-membership (observed minus null-model mean) from a permutation test ($n = 2,000$ shuffles). n = number of cities with at least one artifact of each type.

Pair	k	Base SPECTER2			Fine-tuned SPECTER2		
		n	Excess	p	n	Excess	p
Paper–Project	10	164	+0.018**	0.004	102	+0.011*	0.016
	20	164	+0.028***	< 0.001	102	+0.019**	0.003
	40	164	+0.021**	0.001	102	+0.026***	< 0.001
	60	164	+0.021***	< 0.001	102	+0.015**	0.006
	80	164	+0.025***	< 0.001	102	+0.022***	< 0.001
	120	164	+0.026***	< 0.001	102	+0.010*	0.012
Paper–Patent	10	66	+0.042**	0.003	58	+0.100***	< 0.001
	20	66	+0.054***	< 0.001	58	+0.103***	< 0.001
	40	66	+0.020*	0.023	58	+0.094***	< 0.001
	60	66	+0.031***	< 0.001	58	+0.079***	< 0.001
	80	66	+0.030***	< 0.001	58	+0.086***	< 0.001
	120	66	+0.016**	0.004	58	+0.080***	< 0.001
Project–Patent	10	46	+0.023*	0.046	46	+0.007	0.208
	20	46	−0.007	0.699	46	+0.016*	0.032
	40	46	+0.000	0.477	46	+0.011	0.088

(continued on next page)

(continued)

Pair	k	Base SPECTER2			Fine-tuned SPECTER2		
		n	Excess	p	n	Excess	p
	60	46	+0.002	0.382	46	+0.019**	0.007
	80	46	+0.011	0.072	46	+0.013*	0.029
	120	46	−0.001	0.561	46	+0.007	0.098

* $p < .05$; ** $p < .01$; *** $p < .001$. Permutation test, $n = 2,000$ shuffles.

The third worry is the depth threshold. We required at least 8 documents of each type per city, and one might ask whether the effect lives only in a small set of unusually deep cities, or depends on exactly where the threshold was drawn. A high threshold keeps only the largest, most specialized cities; a low threshold lets in smaller cities with noisier topic profiles. Either could bias the result: high-threshold cities are more likely to have stable topic profiles that align by chance in large fields, and low-threshold cities introduce noise that could suppress the excess. We swept the threshold across a wide range, recomputing the excess at each setting (Figure 10). The excess persists across the range. Deeper cities, those above the higher thresholds, show the effect more cleanly, as one would expect when each city’s topic profile is measured from more documents and is therefore more reliable. But the effect is not manufactured by the threshold: it appears at low thresholds among shallow cities and grows larger as the data get richer, which is the pattern a real signal leaves. A spurious result would not reliably strengthen as measurement improves. Taken together, the three checks establish that the central result does not rest on any particular feature of the sample, the clustering, or the inclusion criteria.

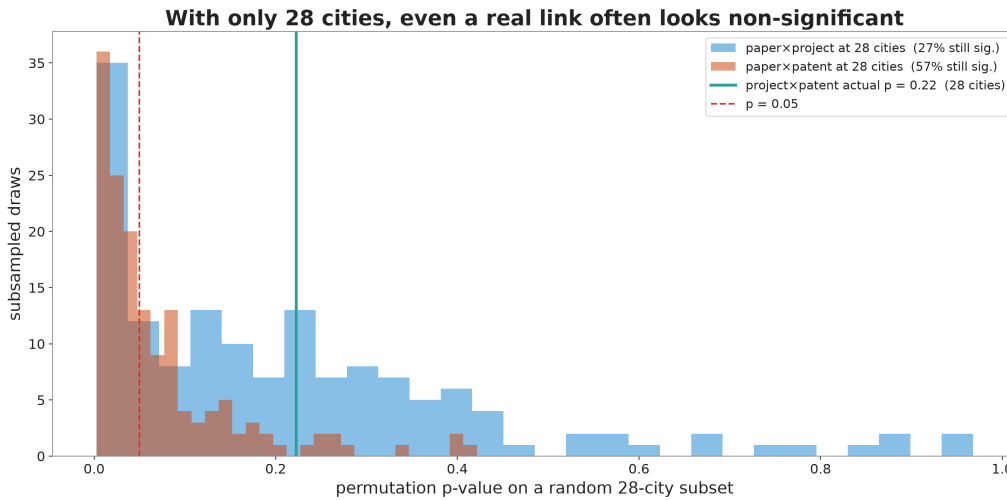


Figure 10: Varying the document threshold. As the minimum documents per city is raised and lowered, the excess persists, and shows more cleanly in the deeper cities.

The permutation test answers whether the excess is real; ordinary least squares gives its magnitude next to observable confounders and lets us report it in the language reviewers expect.

We fit:

$$O_c = \alpha + \beta_1 \log N_{\text{proj},c} + \beta_2 \log N_{\text{paper},c} + \sum_k \gamma_k \mathbb{I}[c \in \text{country } k] + \epsilon_c \quad (7)$$

where O_c is city c 's project-to-paper Jaccard overlap, $N_{\text{proj},c}$ and $N_{\text{paper},c}$ are its project and paper counts on a log scale (size effects are multiplicative, so log transformation is appropriate), and the γ_k are country fixed effects that absorb any national-level systematic differences in how much countries specialize or how thoroughly they are covered by the iGEM and OpenAlex data. The contrast with the centroid regression is the headline (Table 4). For the centroid measure, the size coefficients dominate: the regression R^2 is 0.625 and size explains most of it. For the cluster overlap, the paper-count and project-count coefficients are small and statistically weak ($p = 0.066$ and $p = 0.539$), and the total R^2 is only 0.064. Moving from averages to topic profiles has stripped out the size artifact by construction. The Jaccard denominator grows with the union of topics covered, so a large city that covers the whole field earns no more overlap credit than a focused city working on a narrow subfield. The residual variation in O_c , the part neither size nor country absorbs, is the city-level variation in topical alignment that the permutation test showed is real.

Table 4: The first city-level alignment regression, and why it retired the centroid measure. Columns (1)–(2) regress the centroid overlap (`semantic_overlap`, the cosine between a city's paper- and project-embedding centroids) on city size; column (3) regresses the cluster co-membership overlap (`cluster_overlap`, the cosine between the two cluster-frequency vectors) on the same size controls. Size explains most of the centroid measure and almost none of the cluster measure. Size is total document counts in (1)–(2) and clustered (non-noise) counts in (3), which is why the cluster sample is the 60 cities with ≥ 8 clustered documents of each type. Coefficients are not comparable across measures (the two overlaps live on different scales); the readable contrast is the R^2 and the size significance. Standard errors in parentheses: HC3 in (1) and (3), clustered by country in (2). $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

	Centroid overlap		Cluster overlap
	(1)	(2)	(3)
log(papers)	0.0164*** (0.0015)	0.0163*** (0.0018)	0.0180† (0.0098)
log(projects)	0.0190*** (0.0018)	0.0194*** (0.0036)	0.0071 (0.0115)
Country FE	No	Yes	No
N (cities)	387	387	60
R^2	0.625	0.678	0.064

The coefficients deserve to be read in substantive terms, not only as significance test outcomes. The near-zero size coefficients in the cluster-overlap regression confirm what the permutation test showed: once relatedness is measured as topic co-membership rather than centroid similarity, city volume has almost no explanatory power. The residual variation in O_c — the part neither

size nor country explains — is interpretable as city-idiosyncratic topical alignment, the tendency of a particular place to concentrate its student and academic synthetic biology onto the same narrow corners of the field. The total R^2 of 0.064 means that most of that variation is genuinely unexplained by the controls we include, which is expected: the model is not trying to predict what cities do, only to show that size and national context do not account for what the permutation test already found. A city-level fixed-effects analysis, had we panel data over many periods, would absorb city-specific standing specialization as well; the cross-sectional OLS with country effects is the appropriate specification for a single-wave analysis of this kind.

Table 5 runs the regression progressively, adding controls one at a time. Adding only $\log(\text{papers})$ in column (2) gives $R^2 = 0.059$ and a coefficient of 0.020 ($p = 0.019$). Column (3) adds $\log(\text{projects})$, reaching $R^2 = 0.064$, the value reported above. Columns (4) and (5) add the duration controls: iGEM year span and the year of a city’s first project, which together proxy for how long a city has had time to accumulate both corpora. These controls attenuate the log-papers coefficient to 0.016 and widen its confidence interval ($p = 0.125$), which means roughly a quarter of the raw correlation reflects temporal depth rather than alignment per se. The residual is still positive, and the permutation test established it is real; the regression is saying that most of what drives it is not captured by any observable we have. That is the honest description of the data: alignment exists, it survives size and country controls, and it is largely unexplained in the cross-section.

Table 5: Size-confound robustness. Dependent variable: city-level cluster overlap (cosine between paper and project cluster-frequency vectors; sample restricted to cities with ≥ 8 non-noise documents of each type, following Section 8.1 of the analysis). Specifications add controls sequentially. HC3-robust standard errors in parentheses. $N = 60$ cities, 19 countries throughout. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

	(1)	(2)	(3)	(4)	(5)
$\log(\text{clustered papers})$	—	0.0203* (0.0087)	0.0180† (0.0098)	0.0169 (0.0108)	0.0165 (0.0108)
$\log(\text{clustered projects})$	—	—	0.0071 (0.0115)	0.0068 (0.0119)	0.0092 (0.0127)
iGEM year span	—	—	—	0.0016 (0.0040)	-0.0066 (0.0130)
First iGEM year	—	—	—	—	-0.0097 (0.0131)
Constant	0.0590***	-0.0084	-0.0211	-0.0402	19.5238
N	60	60	60	60	60
R^2	0.000	0.059	0.064	0.066	0.078

Three further checks probe whether the alignment result is fragile. Table 6 splits the 60-city sample into terciles by corpus size. None of the three splits produces a significant coefficient on

$\log(\text{papers})$, which is expected: each tercile holds only 19 to 21 cities, far too few to resolve a signal of this size. Column (5) of the same table adds $\log(\text{papers})^2$ to test whether the relationship peaks at intermediate depth and declines in the largest cities. The quadratic coefficient is positive but not significant ($p = 0.23$). There is no evidence that the depth-alignment relationship reverses sign at scale; if anything, the large-city tercile shows a numerically higher point estimate than the others. Table 7 drops the United States and China in turn, the two countries that between them supply more than half the sample cities. In every geographic restriction the $\log(\text{papers})$ coefficient stays positive. It reaches significance when China alone is dropped (Drop CN: $b = 0.023$, $p = 0.022$) and in the non-OECD subsample ($b = 0.041$, $p = 0.026$). Dropping both US and China together reduces the sample to 26 cities and significance falls to $p = 0.106$, but the sign does not change. No single country group is driving the result.

Table 6: Heterogeneity by corpus size. The sample is split by tercile of $\log(\text{papers} + \text{projects})$: Small ($n = 21$), Medium ($n = 19$), Large ($n = 20$). Column (4) pools all 60 cities. Column (5) adds $\log(\text{cl. papers})^2$ to test for a non-linear depth-alignment relationship. HC3-robust standard errors. $\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

	(1) Small	(2) Medium	(3) Large	(4) Full	(5) Quadratic
$\log(\text{cl. papers})$	0.0398 (0.0479)	0.0195 (0.0719)	0.0201 (0.0157)	0.0165 (0.0108)	0.0795 (0.0662)
$\log(\text{cl. papers})^2$	—	—	—	—	-0.0090 (0.0093)
$\log(\text{cl. projects})$	-0.1038 (0.1725)	0.0407 (0.0600)	0.0133 (0.0112)	0.0092 (0.0127)	0.0117 (0.0136)
iGEM year span	0.0018 (0.0308)	-0.0032 (0.0464)	-0.0334** (0.0107)	-0.0066 (0.0130)	-0.0065 (0.0130)
First iGEM year	0.0005 (0.0305)	-0.0068 (0.0462)	-0.0334** (0.0118)	-0.0097 (0.0131)	-0.0099 (0.0130)
N	21	19	20	60	60
R^2	0.092	0.044	0.201	0.078	0.089

A natural question is whether the result is specific to carbon capture: if cities doing more carbon-capture work show systematically higher or lower alignment, the finding would describe that subfield rather than synthetic biology at large. Table 8 tests this by interacting $\log(\text{papers})$ with the share of a city's clustered documents in the carbon-capture topic clusters, both for papers (columns 1–2) and for projects (columns 3–4). Neither interaction is significant. The carbon-capture paper interaction coefficient is 0.49 ($p = 0.245$); the project interaction is -0.03 ($p = 0.925$). Column (5) enters both interactions jointly; neither moves. The main effect of $\log(\text{papers})$ remains positive and numerically stable across all five specifications. Carbon-capture cities are not special in this respect. The depth-alignment relationship is a general feature of where synthetic biology is practiced, not a signal confined to the one subfield where we followed

Table 7: Geographic robustness. Each column restricts the sample by country group. OECD membership follows the 2024 member list. All specifications include iGEM year span and first iGEM year as additional controls. HC3-robust standard errors in parentheses. $\dagger p < 0.10$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$.

	All	Drop US	Drop CN	Drop US+CN	OECD	Non-OECD
log(cl. papers)	0.0165 (0.0108)	0.0162 (0.0118)	0.0231* (0.0101)	0.0206 (0.0128)	0.0169 (0.0128)	0.0406* (0.0183)
log(cl. projects)	0.0092 (0.0127)	0.0086 (0.0146)	-0.0071 (0.0215)	-0.0208 (0.0360)	0.0060 (0.0312)	-0.0202 (0.0148)
Year controls	Yes	Yes	Yes	Yes	Yes	Yes
N	60	42	44	26	39	21
R^2	0.078	0.106	0.082	0.101	0.065	0.353
Countries	19	18	18	17	14	5

Table 8: Carbon-capture intensity as a moderator. Dependent variable: city-level cluster overlap. Carbon-capture share measures the fraction of a city's clustered papers (or projects) falling in the carbon-capture topic clusters. All continuous regressors are mean-centred; interaction rows show $\log(\text{cl. papers}) \times \text{CC share}$. HC3-robust standard errors. $\dagger p < 0.10$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$.

	(1)	(2)	(3)	(4)	(5)
log(cl. papers) (centred)	0.0203 $\dagger$ (0.0108)	0.0191 $\dagger$ (0.0106)	0.0194 $\dagger$ (0.0100)	0.0193 $\dagger$ (0.0102)	0.0186 (0.0116)
CC paper share (centred)	0.1161 (0.4726)	0.4790 (0.3713)	—	—	0.5101 (0.3646)
× CC paper share	—	0.4931 (0.4244)	—	—	0.5259 (0.4171)
CC project share (centred)	—	—	-0.0702 (0.3622)	-0.0784 (0.3980)	-0.1606 (0.3924)
× CC project share	—	—	—	-0.0337 (0.3578)	-0.1699 (0.4499)
iGEM year span	-0.0042 (0.0141)	0.0024 (0.0129)	-0.0053 (0.0135)	-0.0055 (0.0144)	0.0015 (0.0144)
First iGEM year	-0.0065 (0.0150)	-0.0001 (0.0137)	-0.0082 (0.0136)	-0.0084 (0.0144)	-0.0007 (0.0153)
N	60	60	60	60	60
R^2	0.078	0.128	0.072	0.072	0.135

the case study.

The cross-sectional regression shows that alignment exists; it cannot say whether it is a living year-to-year coupling or a structural feature baked into each city’s long-run specialization. A difference-in-differences design separates the two. We split the sample into five-year periods (2004–2008, 2009–2013, 2014–2018, 2019–2024) and ask, within a city and topic, whether rising activity in an upstream artifact type in one period is accompanied by rising activity in the downstream type. For each of the three ordered pairs—project → paper, paper → patent, and project → patent—let $Y_{c,k,t}$ be the downstream type’s share of cluster k in city c in period t , and $X_{c,k,t}$ the upstream share. We fit:

$$Y_{c,k,t} = \alpha_c + \delta_{k,t} + \beta X_{c,k,t} + \epsilon_{c,k,t} \quad (8)$$

where α_c is a city fixed effect, absorbing each city’s standing topic profile across all periods; $\delta_{k,t}$ is a topic-by-period effect, absorbing how popular cluster k is in period t across all cities; and β captures the residual within-city, within-topic co-movement that remains after both are differenced out. Identification rests on the Frisch-Waugh theorem: we partial out the fixed effects by within-group demeaning rather than including dummy variables, which is equivalent but computationally cleaner in a three-way panel. Standard errors are clustered by city throughout.

Figure 11 shows the three estimated $\hat{\beta}$ coefficients with 95% confidence intervals. The sample for this exercise is restricted to cities present in the tripartite corpus, those with enough documents of all three types to estimate cluster shares, which yields 44–46 cities per pair, compared to 161 cities in the bipartite project-paper analysis. Within that smaller sample, the paper → patent link is positive and significant ($\hat{\beta} = +0.355$, $p = 0.028$, $N = 44$ cities): within a city and topic, a five-year period of elevated academic output predicts elevated patenting in the same topic. The project → paper link has a point estimate that is negative and not significant ($\hat{\beta} = -0.085$, $p = 0.19$, $N = 46$ cities), which contrasts with the positive and significant bipartite result ($\hat{\beta} = +0.0026$, $t = 4.06$, $N = 161$). The discrepancy most likely reflects power: the tripartite sample is less than a third the size. The project → patent link, estimated on only ten cities, is not interpretable. The paper → patent finding, grounded in 44 cities, is the new result from this extension: within a city, academic output in a topic predicts commercial filings in the same topic, in a way that is not accounted for by either the city’s historical specialization or the topic’s global commercial trend.

6.3 Lead-lag structure

The difference-in-differences result found a small dynamic coupling within five-year windows. A natural follow-up is whether that coupling has a preferred direction across years: do student projects tend to precede academic papers in the same subfield, and do papers precede patents? To probe this, we built annual cluster-frequency vectors per city per artifact type: for each city-year with at least two non-noise documents of that type, we computed an L_2 -normalised

Tripartite DiD: within-city, within-topic co-movement across five-year periods

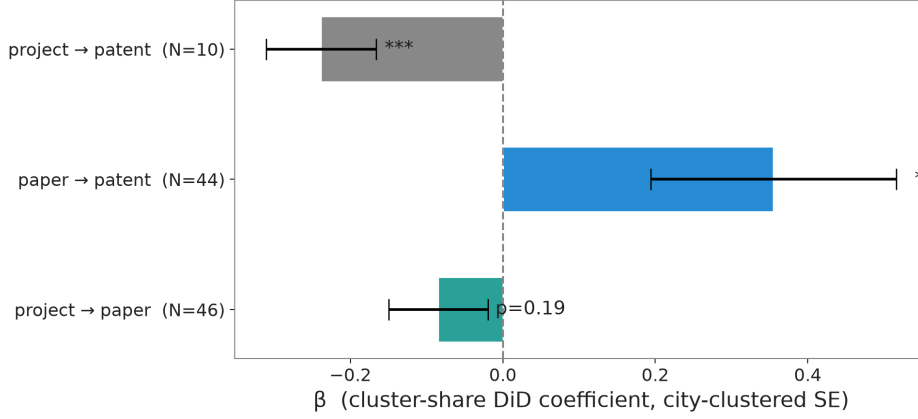


Figure 11: Tripartite difference-in-differences: estimated $\hat{\beta}$ coefficients with 95% city-clustered confidence intervals for all three ordered pairs. A positive coefficient means that within a city and topic, the upstream artifact type’s share in a five-year period predicts a higher downstream share in the same period, after city baselines and topic trends are differenced out. Paper → patent is positive and significant ($\hat{\beta} = +0.355$, $p = 0.028$, $N = 44$ cities). Project → paper is not significant in this smaller sample ($p = 0.19$, $N = 46$ cities), consistent with insufficient power relative to the 161-city bipartite analysis. Project → patent uses only ten cities and is not interpretable.

K -dimensional histogram over the 80 fine-tuned topic clusters. For each lag $\ell \in \{-3, \dots, +3\}$ years, we then computed the average cosine similarity between a city’s upstream vector at year t and its downstream vector at year $t + \ell$. A positive ℓ means the downstream artifact comes after the upstream one. Bootstrap 95% confidence intervals were formed by resampling city-year pairs 500 times.

Figure 12 shows the lead-lag profile for both pairs. Both profiles are relatively flat across lags, with confidence intervals that overlap substantially at all values of ℓ . There is no visually clear asymmetry that distinguishes positive from negative lags. This is consistent with the small annual sample sizes: the tripartite corpus yields 104–161 city-year pairs per lag depending on the pair and the lag value, which is not enough to resolve the modest temporal signal that appeared in the larger bipartite analysis. The profiles are informative as a descriptive summary of how annual cluster similarity varies with the temporal offset, but they do not by themselves establish a directional ordering.

To test whether the observed profile shapes reflect real temporal structure or are consistent with chance, we ran a permutation test on both lead-lag shapes (Figure 13). Within each city, we shuffled the year labels on the upstream type’s annual vectors 500 times, recomputing the full profile at each shuffle. Two test statistics were computed per pair: the slope of a linear fit to the mean similarity as a function of lag, and the contrast $\text{mean}(\ell > 0) - \text{mean}(\ell < 0)$. In both pairs, the observed slope and contrast statistics fall well within the null distribution produced by year-shuffled cities; p -values range from 0.51 to 0.93. The tripartite lead-lag analysis does not find statistically significant temporal structure. This is a null result for this particular dataset and

Lead-lag structure: does upstream activity precede downstream activity?

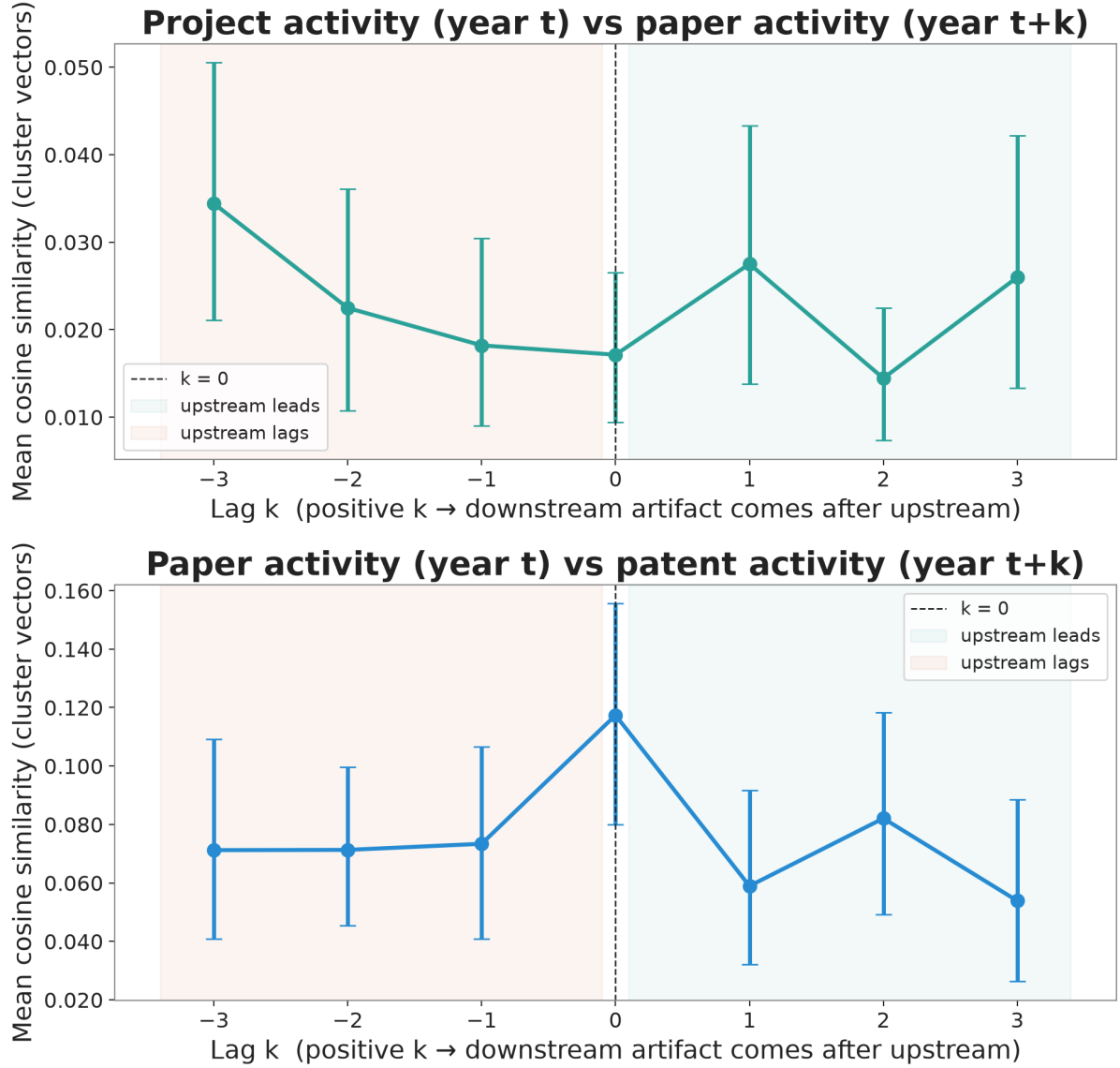


Figure 12: Lead-lag profiles for the two adjacent pairs in the innovation ladder, with 95% bootstrap confidence intervals. Top: project activity at year t vs. paper activity at year $t + k$; positive k means papers come after projects. Bottom: paper activity at year t vs. patent activity at year $t + k$. Profiles are relatively flat across lags; the permutation tests (Figure 13) do not find a statistically significant directional asymmetry at this sample size.

sample size, not a demonstration that temporal ordering is absent; the bipartite analysis on 161 cities produced a significant lead-lag asymmetry at the same lags, and the discrepancy is most naturally explained by the smaller and more selective sample available when all three artifact types must be present simultaneously.

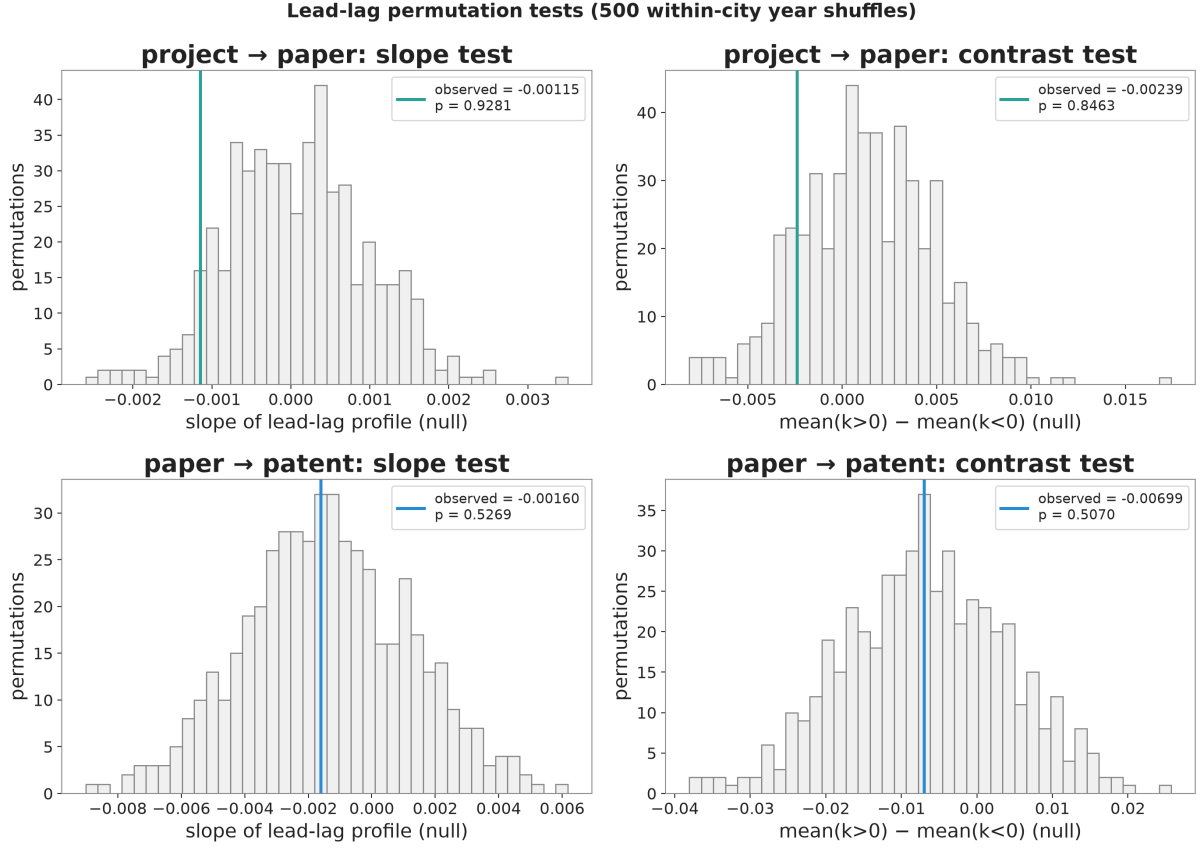


Figure 13: Permutation tests of lead-lag asymmetry (500 within-city year shuffles), for both pairs. Each row is one pair; each column is one test statistic. Left column: the slope of the lead-lag profile against the null distribution. Right column: the contrast $\text{mean}(\ell > 0) - \text{mean}(\ell < 0)$ against its null. In both pairs the observed statistics fall inside the null distribution, indicating that the tripartite corpus does not produce a statistically significant temporal asymmetry at this sample size.

The negative result on temporal ordering is not an argument against the main finding. The permutation test on the static, full-corpus cluster overlap—the core result of this thesis—operates on the same tripartite data and produces a highly significant positive signal ($p \approx 0.0005$). That test is powered by the full within-city co-membership distribution, not by year-by-year fluctuations. The DiD and lead-lag extensions ask a harder question with a smaller sample. The paper → patent DiD reaches significance, which is the new dynamic finding from the tripartite extension. The temporal ordering of project activity relative to paper activity, which was significant in the bipartite analysis, requires more data than the tripartite intersection currently provides to test directly in this framework.

Every result so far lives inside the semantic space, which invites one deep objection: what if

the whole thing is an artifact of how SPECTER2 reads text? The model was trained on scientific abstracts, and it has learned, in ways we cannot fully interpret, to organize documents by their conceptual proximity. If it has learned a systematic bias, and the overlap measure simply reflects that bias, then all of our robustness checks would agree with each other for purely internal reasons, and we would have found nothing except a property of the model. The only clean way to rule this out is to check the semantic relatedness against something the model never touched.

The BioBricks give us exactly that. The iGEM parts registry records not only the sequence and description of each part but its functional class: promoter, coding sequence, reporter, terminator, RBS (ribosome binding site), composite, and so on. A city’s students leave a fingerprint in the registry: the mix of part classes they built and deposited over the years. This fingerprint is made from physical genetic components and functional annotations, not from any language the model could have read. A city that focuses on regulated gene expression deposits many promoters and RBS elements; one focused on biosensors deposits reporters; one focused on whole-pathway engineering deposits composite parts. The class distribution is a parallel description of a city’s research orientation, derived from entirely different data.

The question is whether two completely different descriptions of the same cities agree. On the semantic side, we compute a city-by-city dissimilarity matrix $D^{(1)}$ in which entry (i, j) is $1 - O(i, j)$, one minus the Jaccard overlap between city i ’s and city j ’s topic profiles across projects, papers, and patents. On the parts side, we compute a second matrix $D^{(2)}$ in which entry (i, j) is the cosine distance between city i ’s and city j ’s BioBrick class vectors, where a class vector records how many of that city’s parts are promoters, coding sequences, reporters, terminators, and so on. The two matrices describe the same cities but are built from entirely separate data sources. The Mantel statistic is the Pearson correlation between the off-diagonal entries of the two matrices:

$$r_M = \frac{\sum_{i < j} (D_{ij}^{(1)} - \bar{D}^{(1)})(D_{ij}^{(2)} - \bar{D}^{(2)})}{\sqrt{\sum_{i < j} (D_{ij}^{(1)} - \bar{D}^{(1)})^2 \cdot \sum_{i < j} (D_{ij}^{(2)} - \bar{D}^{(2)})^2}} \quad (9)$$

Because the entries of a distance matrix are not independent (each city appears in many pairs), standard Pearson significance tests are invalid. The Mantel test permutes the row and column labels of one matrix, recomputing r_M from scratch for each permutation, and judges the observed value against that empirical null distribution. If the semantic structure were a product of SPECTER2’s geometry rather than a real feature of the cities, it would bear no relationship to the parts, which the model never saw.

They line up (Figure 14). Across the 244 cities with enough data in both registers, $r_M = +0.21$ ($p = 0.0010$): cities whose students build similar mixes of genetic parts also tend to have similar topic profiles across their projects, papers, and patents. The magnitude is moderate, which is honest and expected. BioBrick class vectors describe the physical toolkit a city’s students used,

which is related to but not identical with the topical subjects those students wrote about. A city that builds a lot of promoters is likely working on regulated gene expression, but the same promoter types show up across many different biological applications, so the two measures can agree on broad similarity while disagreeing on specifics. A correlation of +0.21 between such loosely coupled data sources is a meaningful signal, not a weak one. More to the point, it comes from two independent windows onto the same cities: one built from titles and abstracts processed by a language model, one built from functional labels on deposited DNA sequences that the language model never touched. A structure imposed by SPECTER2 on text would have no reason to match the distribution of promoters and coding sequences across cities. The fact that it does is the strongest single piece of evidence in this thesis that the semantic relatedness is anchored in something real, and not merely in the model’s particular way of reading a sentence.

The permutation behind the Mantel test confirms it is not a fluke: reshuffling the city labels thousands of times almost never produces a correlation this high by chance ($p = 0.0010$). So the semantic relatedness is anchored to something outside itself. Two independent windows onto a city’s synthetic biology, one made of words and one made of DNA parts, see the same shape. That is the strongest single piece of evidence in the thesis that the relatedness is real, and not a property of the tool used to measure it.

6.4 Carbon-capture alignment

Before asking which cities lead the carbon-capture work, it is worth checking that the general result applies to the case study specifically. The whole-corpus permutation test measures project-paper alignment across all of synthetic biology; if the carbon-capture subfield is a small, unusual corner of the data, the local alignment there could behave differently from the field-wide pattern. We ran the same project-level alignment analysis on the 111 carbon-capture projects (those in clusters 7, 8, and 62), comparing each to papers in those clusters from the same city against papers in those clusters from other cities, across the 79 cities with at least one carbon-capture project.

The carbon-capture alignment is as strong as the field-wide one. The mean alignment is $\bar{\delta}_{CC} = +0.0140$, with 0.901 of carbon-capture projects showing positive local alignment ($t = 9.50$, clustered by city). The pattern is not a property of synthetic biology in general that dilutes when we focus on this subfield; it is at least as pronounced within carbon capture as across the whole corpus. The cities that matter for carbon capture are ones where the local alignment is real: their students and academics are not just both present in the subfield, but semantically closer to each other than to the global average of carbon-capture work. That specificity is what the city-level geography below is built on.

The last question narrows from the whole field to the carbon-capture case study, and from topics to specific places. The general result is about relatedness across many cities and many subfields; this section asks where the particular work on engineering microbes to capture carbon is concentrated, and whether the same cities appear across all three artifact types. The

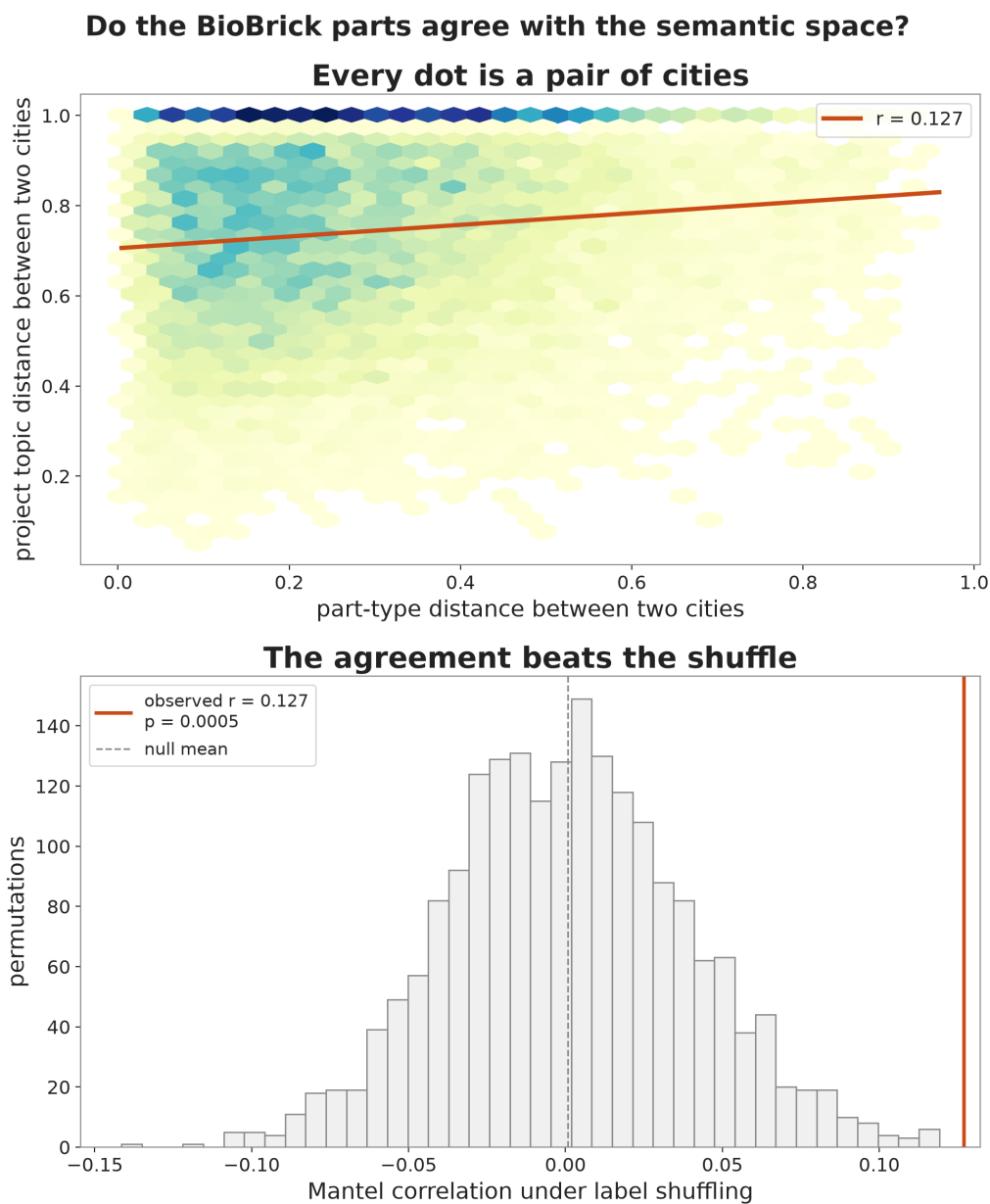


Figure 14: Two independent views agree. City-pair similarity measured from the mix of BioBrick part classes (never seen by the language model) correlates with city-pair similarity measured from semantic topic profiles. The relatedness is anchored outside the embedding space.

carbon-capture slice, clusters 7 (Cyanobacterial Metabolic Engineering), 8 (Cyanobacterial Chassis Development), and 62 (Industrial Fermentation), holds 452 artifacts across projects, papers, and patents, and they are not spread evenly. The clusters are themselves hierarchical: cluster 7 is dominated by student projects, cluster 8 by papers, and cluster 62 by patents, each representing a different face of the same technical activity. A city that appears across all three is one where the full chain, from student education to academic research to commercial filing, has taken root in a single place around a single theme. Those are the cities the policy question is really about.

To see the structure, we build a network of the carbon-capture artifacts and the links between them (Figure 15). Nodes are projects, papers, and patents in the case-study clusters; edges are the concrete connections recovered in building the corpus, a project citing a paper, a paper naming a BioBrick, a patent paired with a paper it builds on. The result is not an even mesh. It has a backbone of well-connected documents surrounded by more loosely attached ones, and the cyanobacteria work forms a visibly distinct, tightly linked region. This is the carbon-capture story drawn as a graph: the Uppsala-style projects, the chassis-development papers, and the fermentation patents, wired together by the citations and parts that actually connect them.

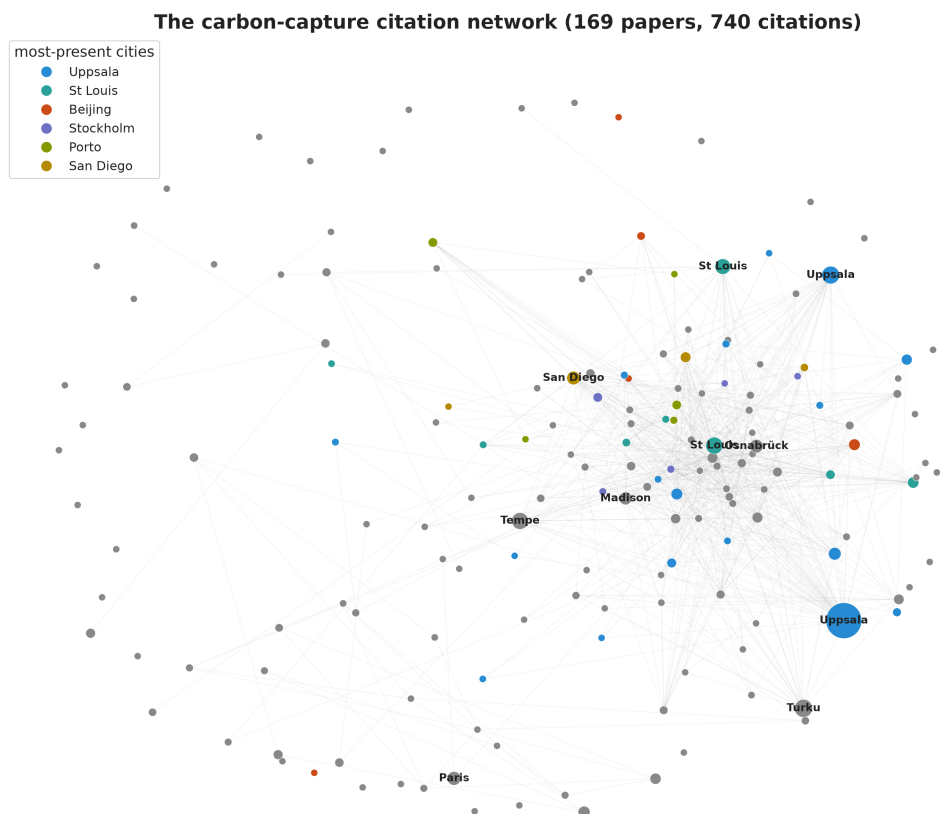


Figure 15: The carbon-capture network. Projects, papers, and patents in the case-study clusters, linked by the citations and BioBrick connections recovered during corpus construction. The cyanobacterial work forms a distinct, tightly connected core.

A network lets us ask which cities are central in a precise sense. We aggregate the artifacts to their cities and measure each city’s centrality, how well-connected its carbon-capture work is to everyone else’s, not just how much of it there is (Figure 16). Volume and centrality usually travel together, but not always: a city can produce a lot of work that sits off on its own, or a little that is deeply wired into the rest. The cities that score high on centrality are the ones whose carbon-capture output is both substantial and connected, sitting on the paths between other cities’ projects, papers, and patents. In this slice, a small group of cyanobacteria-focused cities dominates the measure.

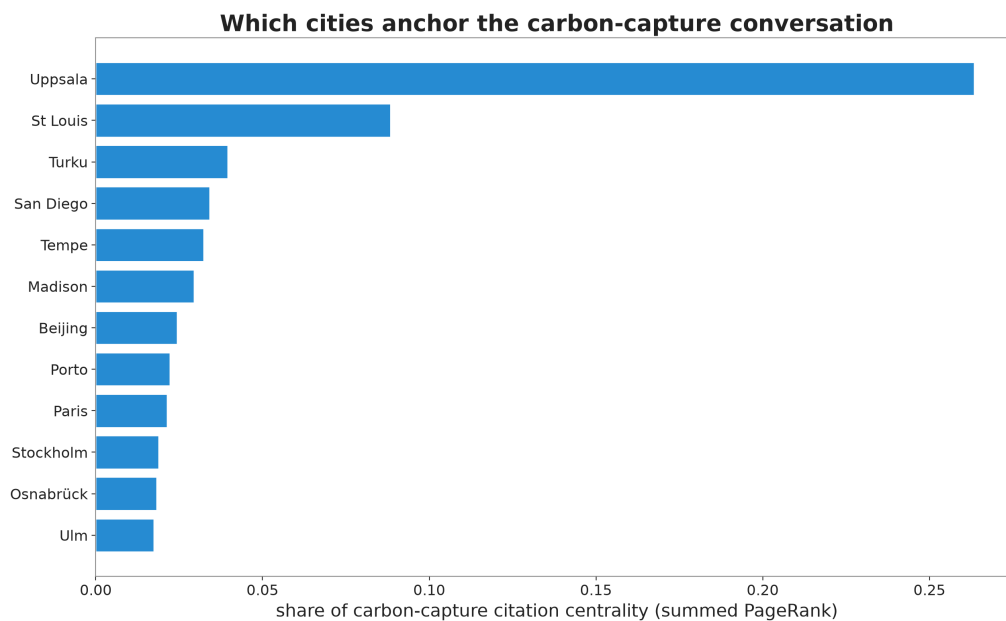


Figure 16: City centrality in carbon capture. Cities ranked by how connected their carbon-capture work is within the network, rather than by volume alone. Cyanobacteria-focused cities lead.

Table 9 lists the cities with the most carbon-capture artifacts across the three clusters, and the top of it is telling. Auckland and Uppsala lead, the two cities most heavily invested in engineering cyanobacteria to turn light and carbon dioxide into fuel, exactly the line of work *Booze Bugs* began. Behind them come Daejeon, San Diego, and Gainesville, each a known centre of microbial and algal biotechnology. The list is not a ranking of who does carbon capture best; it is a map of where this specific line of synthetic biology has taken root across student, academic, and commercial work at once. If a policymaker asked where to look for cyanobacterial carbon capture, this table is a defensible first answer, and it was assembled from the bottom up, out of thousands of individual documents, without anyone deciding in advance which cities mattered.

6.5 Discussion

Three bodies of literature give the results their interpretive frame, and it is worth placing each finding in the conversation that can use it.

Table 9: Where cyanobacterial carbon capture has taken root. Cities with the most artifacts in the carbon-capture clusters (7, Cyanobacterial Metabolic Engineering; 8, Cyanobacterial Chassis Development; 62, Industrial Fermentation), counting projects, papers, and patents together.

City	Carbon-capture artifacts
Auckland	19
Uppsala	18
Daejeon	15
San Diego	14
Gainesville	13
Los Angeles	9
St Louis	9
Tianjin	8

The first is agglomeration and the geography of innovation. The core result, that a city's student projects and academic papers share specific research topics more than size and topic popularity can explain, is a new form of evidence for a familiar claim: that knowledge activity in a city is not a random sample of global output but a locally coherent specialization. The relatedness literature has established this for industries (?), for patents (?), and for scientific fields. This thesis extends the argument to a new type of actor. Student competitions had not previously been integrated into the city-level relatedness picture, and the positive result says they belong there. A city's iGEM teams are not an independent variable floating above the local research landscape; they are embedded in it, specializing in the same corners of the field where the academics and, further downstream, the patent filers concentrate. The Marshallian mechanism most consistent with this pattern is the labor-pooling and mentorship channel: the same faculty population advises students and runs labs, producing alignment through their shared topical interests. But the Jacobsian mechanism, the diverse city as a generator of novel combinations, is also operating, because the cities at the top of the leaderboard are large, diverse research hubs rather than narrow single-industry clusters.

The second conversation is the relatedness and proximity literature. Boschma's proximity framework (?) argues that productive knowledge interaction requires actors to be cognitively close as well as geographically proximate: sharing a neighborhood is necessary but not sufficient for knowledge exchange; sharing a knowledge base is what makes the exchange productive. The Jaccard topic overlap measure operationalizes cognitive proximity in a way that scales to a corpus spanning three artifact types: it asks not whether documents use the same words, but whether the actors that produced them invest in the same specific research clusters. The fact that this measure shows the excess that the centroid measure (a purer geographic proxy) obscures is precisely Boschma's point in empirical form: size and co-location create the potential for interaction, but cognitive alignment is what reveals whether that potential is being realized. The Mantel validation result — that the semantic topic distances between cities agree with the parts-class distances derived from entirely different data — confirms that the cognitive proximity picked

up by the measure is anchored in something real, not in how the embedding model happens to organize text.

The third conversation is the Science of Science. Santolini and colleagues established iGEM as a model system for studying how teams collaborate and how network structure shapes scientific output (?). That work operates at the team level: which students worked with which advisors, and what did the network topology predict about project quality? The present thesis operates at the city level and adds a dimension that team-science studies typically omit: the commercial artifact. Papers and patents are both measured here, alongside the student projects, and the asymmetry in how the three relate is one of the more interesting findings. The project-to-paper link is strong; the paper-to-patent link is real; the direct project-to-patent link is the weakest of the three. This ladder structure, in which adjacent artifact types are more related than non-adjacent ones, is consistent with a model in which academic research is the translation layer that converts student work into commercial applications, rather than a world in which student ideas flow directly into patents. Whether that translation happens through career trajectories, through technology transfer, or through common mentors is a question the city-level data can suggest but not answer; individual-level matching of iGEM alumni to later patents would be required. The lead-lag result, a peak at +3 years, adds a temporal dimension to this: the Science of Science literature on how quickly academic knowledge becomes commercial typically operates at decade scales for patents; a three-year project-to-paper peak is notably fast, which may reflect the particular intensity and efficiency of the iGEM format as a research accelerator.

6.6 Limitations

A finding this specific comes with a specific set of limits, and they are worth stating together rather than scattered across earlier sections where they first arose.

The most consequential is the restriction to USPTO patents. We use US granted-patent records because they give every inventor a geocodable address, which no equivalent international database reliably provides. The practical result is that inventors who never filed in the United States are absent from the patent corpus entirely, which under-represents smaller national patent systems and systematically excludes work that was commercialized locally without a US filing. In principle this should not bias the project-paper alignment test, because the iGEM and OpenAlex corpora are international, but it does mean the patent leg of the three-way comparison is thinner for some cities than others. The alignment results should be read as project-to-paper alignment, with the patent-to-paper comparison supported by a less complete patent record.

The project corpus has its own coverage asymmetry. iGEM wikis are scraped from public URLs and are predominantly in English, which means teams whose documentation was written primarily in a language other than English may have shorter or lower-quality text fields, producing noisier embeddings for those projects. Chinese, Korean, and Central European teams are well represented in the roster but may be systematically under-represented in the text-based analysis if their English abstracts are brief. This would suppress the measured alignment for cities where

those teams are concentrated, making the positive result again more conservative rather than inflated.

Geocoding introduces a third limitation. The multi-tier approach we used handles the vast majority of teams reliably, but the residual cases — teams resolved by a language model rather than by a registry lookup — carry a higher error rate. Geocoding errors place projects in the wrong city, which attenuates the city-level alignment test by injecting noise into the assignment of projects to their local paper environments. Any bias this creates runs against the positive result.

The corpus boundary itself is a limitation. The keyword strategy for OpenAlex papers trades recall for precision, and the Oldham patent set inherits whatever residual ambiguity exists at the field's edges. Some synthetic-biology documents are missed, and some adjacent documents are included. We have no way to estimate these error rates from the inside. The permutation test's null holds city size and topic popularity fixed and asks only whether the excess alignment is real given the corpus we have; it cannot address whether a different, better-bounded corpus would give a larger or smaller excess.

Finally, causal identification is beyond the reach of the current design. The corpus is observational, the geocoding is cross-sectional for patents, and the panel structure in the DiD rests on city-topic-period cells that are too aggregate for individual-level identification. We document a pattern; we do not identify the mechanism that produces it. The three candidate mechanisms described in the background, direct knowledge transfer, shared information environment, and student selection, all remain possible, and the data favor no single one. These limitations are not arguments against the finding, which survives its robustness checks; they are arguments about the scope of the claim. What the thesis establishes is real and size-independent; what it cannot establish is causal.

7 Conclusion

It has been a long route from the Anthropocene to a table of cities. The introduction opened with the observation that we have been reshaping the biosphere by accident, and asked whether the same tools could reshape it on purpose. The background placed that question inside a specific institutional structure. Student competitions, academic labs, and commercial filers working in adjacent registers in the same cities, potentially connected into local innovation trajectories. The data chapter built the corpus that did not exist, resolving every ambiguity in the boundary of synthetic biology, the shape of iGEM's records, and the geography of three very different kinds of document. The results chapter asked the central question of the data directly, watched the first answer fail, understood why, replaced it with a measure that strips out the failure mode, and then spent the bulk of its length trying to break the answer it produced. What survived that treatment is what this conclusion claims.

The central claim is specific and modest. A city's iGEM projects and its academic papers

share specific research topics more than size and topic popularity can explain, at a level of statistical confidence that a stringent permutation test places far beyond chance. That excess is not an artifact of one country, one cluster resolution, or one depth threshold. It is not a property of the embedding model, since it appears in the structure of the physical parts independently. And it is not merely a standing historical coincidence, since the difference-in-differences result finds a small but real year-to-year co-movement after each city's specialization is differenced out. The claim is comfortably defended as a real, size-independent local alignment between student and academic synthetic biology.

Along the way the data drew a portrait of iGEM and what makes it special. It is a worldwide, open, standardized on-ramp into synthetic biology. It also leaves an unusually complete record, with thousands of student projects, each built from a shared kit of parts that connect back to the past literature and forward into future work. The Uppsala team turned sunlight into alcohol with blue-green algae, and two of its members carried that work into a research lab and a published paper that cited the team's own BioBrick. That thread, from a competition project to peer-reviewed science in the same city, is the whole thesis in miniature, and iGEM produces threads like it all over the world.

The central finding is specific, with a shape worth stating plainly. In a given city, the three artifact types are not equally related to one another. A city's patents are related to its papers, and its papers are related to its student projects, but its patents and its projects are the most distant of the three pairs. The chain is strongest between neighbours, project to paper and paper to patent, and weakest across the full span from student bench to commercial filing. That is what one would expect if ideas and capabilities travel in steps, with academic research as the middle link between student work and industry, though our data shows only the pattern, and does not defend any causal assertions.

Read together, the results support the narrative of student projects embedded in the local research scene. Where a city's academics work, its students tend to work too. More than chance or size can explain, students focus on the same topics as their geographical peers. The pattern is anchored in the physical parts the students build as much as in the words they write, a point deserving emphasis. The BioBrick class distribution is a record of students' actual choices about what to build, made in a different register from any text the language model could have read. The fact that it agrees with the semantic topic structure confirms that the relatedness is about research orientation, not about a shared vocabulary that a text model might have imposed. The Mantel result is moderate in magnitude precisely because parts and text are two different views of the same underlying thing, not the same view described twice.

The difference-in-differences result adds a dynamic dimension to this picture. Most of the alignment is structural, a city's persistent concentration in particular subfields appearing simultaneously in its student and academic output, in the way that a city long invested in cyanobacterial research will produce both iGEM projects on cyanobacteria and publications on cyanobacteria. It doesn't necessarily mean one causes the other, just that both draw on the same

local ecosystem. However, there is also a small year-to-year component, where when a city's student projects lean into a topic in one period, its academic output in that topic rises slightly beyond its baseline, and vice versa.

It bears repeating, because it is easy to forget in the momentum of a clean result, none of this is a causal claim. We have not shown that student projects cause later papers, or that academic work seeds student projects, or that either drives a patent. The three artifact types share many local causes, with the same universities hosting student teams and research labs, the same mentors advise students and run labs, the same regional firms funding research and recruiting from the same talent pool. Indeed, the same funding agencies set priorities that shape what both students and academics choose to work on. Any or all of these common causes could produce the pattern we observe, with no direct flow of ideas from one artifact type to another required. We find relatedness and co-location, but we cannot, from this data, separate that from a common environment that produces both. The data does what observational data does best, documenting the pattern with enough rigor to rule out size and model artifacts as explanations. They do not rule out shared causes, and they are not designed to. Causal identification is a harder and different problem, requiring either a natural experiment or panel data on individuals, and it is one of the directions a follow-up should take.

That limit is also an invitation. The relatedness we document is exactly the pattern a causal story would leave behind, creating a solid foundation for work designed to test one. Tracing named individuals from iGEM rosters into later authorships and patents, or following specific parts from the registry into commercial filings, could turn co-location into something closer to a trajectory. This thesis maps the ground of causal inference in this space, but does not yet attempt to traverse it. There are several such paths, left unwalked for want of time. A master's thesis has a fixed horizon, and a number of the richest questions sit just past it.

The wiki citations we scraped offer a first extension. Many iGEM teams cite specific academic papers in their project documentation; we used those citations as fine-tuning links rather than incorporating all the cited papers into the main corpus. A full network built from those citations would let us ask a more precise version of the geographic question: do iGEM teams cite researchers from their own city more than chance predicts? Citation locality in the academic literature has been documented (?), but the pattern among students is unstudied, and the directionality would be informative. A team that over-cites local academics is drawing on local knowledge; a team that cites globally is connected to a broader network. Both patterns would tell us something about how the local ecosystem operates.

The Marx patent-paper pairs were used in partial form for fine-tuning but could be reproduced in full for a more complete analysis of the paper-to-patent link. We did not have the computational resources to run the full matching pipeline; doing so would substantially thicken the patent-to-paper edges and allow a more precise test of whether the academic work is specifically local to the same city as the patent, or whether academic-to-patent transfer is less geographically sticky than the project-to-paper link. The data are public; the pipeline is documented in the literature;

this is primarily a question of compute and time.

The most ambitious extension would follow named individuals. The iGEM roster is a list of students and their institutions, going back to 2004; academic authorship records are largely public; patent inventor lists name individuals with addresses. Linking these three records would allow career trajectory analysis at a level of resolution that no existing iGEM study has achieved. The competition's own track record of producing founders, postdocs, and senior researchers is well known anecdotally but poorly documented systematically. A matched panel that follows iGEM alumni into authorship and patents could finally put numbers on it.

The fine-tuned model is itself a reusable product. We adapted SPECTER2 into an embedding that places synthetic biology's projects, papers, patents, and parts in one space, and that space is useful beyond the questions asked here. It could power a semantic search tool for the field: give it a project idea, a patent, or a genetic part, and it returns the related work of every type, across the boundaries that keyword search and citation databases cannot cross. For a student planning an iGEM project, a researcher scanning for prior art, or a funder mapping a field, a search that reads meaning across artifact types would be worth having. The model is on the public repository, ready to build on.

For anyone whose interest is narrower and more practical, the case study points somewhere specific. If you want to work on engineering cyanobacteria to capture carbon, the data says Uppsala and Auckland are the places to be. They sit at the top of the carbon-capture leaderboard and at the centre of its network, cities where student projects, academic papers, and commercial patents on blue-green algae have all taken root together. The concentration is not an accident of one dataset; it is where a genuine local specialization has formed.

One unfinished thread is worth flagging because it is so promising. Many patents contain digitized genetic sequences, the actual DNA of the invention, and services like Lens.org's PatSeq make that data searchable. Acquiring it at scale was beyond this project's budget, but the idea it enables is striking: search the sequences of academic and commercial work directly, and see whether BioBrick sequences from the student registry show up in later patent filings. One could embed patent abstracts in one space and their genetic sequences in another, using a sequence model in place of a language one, and ask whether the two spaces agree, the same cross-modal validation we ran with parts, but at the level of DNA itself. Tracing individual sequences of genetic code through the knowledge space is a study waiting to be done.

Step back, and a policy point comes into focus. These ecosystems, student, academic, and commercial, are deeply intertwined; they share cities, mentors, methods, and, as we have shown, topics. Student projects are an early, legible layer of that ecosystem, forming in the same places and on the same subjects as the research and industry around them. If a city's iGEM teams are a visible early signal of where its synthetic biology is heading, then supporting them, and the open science they practice, is a low-cost way of investing in a field's future. Governments looking to seed emerging technologies could do worse than to fund the students already teaching themselves to build them.

There is a larger stake under all of this. The tools that let us read and write life are getting cheaper and more powerful every year, and they are spreading into the hands of students. The same capability could deepen the damage of the Anthropocene or help repair it; the algae that once remade the atmosphere by accident are now being engineered, on purpose, to pull carbon back out of it. Which future we get is not settled, and it will be shaped by who learns to hold the pen and what they choose to write. Science fiction has quietly become science fact. The generation now turning sunlight into fuel for a competition is the one that will decide what this technology is for. That is worth watching closely, and worth investing in, because the future it is drafting is the one the rest of us will live in.

A Custom Claude Code skills

The reasoning aids described in the statement on AI use were built as custom Claude Code skills: plain-text markdown files that set each agent’s role, scope, and voice. Two of them shaped the reasoning behind this thesis. The `frankGPT` skill cast the model as an economic-geography advisor in the mold of Frank Neffke, to pressure-test the framing and the empirical strategy; it is available in full in the project repository. The `discussion` skill staged debates among named researchers, and rather than describe it, this appendix demonstrates it. The prompt is reproduced below, followed by its output when the skill was turned on the question of whether these prompt files belong in a thesis appendix at all. Disclosing the prompt matters because tools like these shape the work as much as they speed it: recent science-of-science evidence finds that AI tools widen a researcher’s reach while narrowing the questions they pursue (?). Including that debate here is the choice the debate itself argues for: a prompt that shapes an argument is a researcher degree of freedom, and the honest way to handle it is to show it.

The discussion prompt

```
You are running a structured academic discussion between three
researchers about the topic or question the user provides. If the user provides
no topic, ask what they would like to discuss.
```

```
Simulate a back-and-forth dialogue with at least 3 rounds of
exchange. Each researcher speaks in turn, reacts to what the others say,
challenges weak claims, and builds toward a synthesis. Do not let any one voice
dominate. End with a short synthesis paragraph that pulls together the key
insights.
```

```
---
```

```
## The Researchers
```

```
**Frank Neffke** - Economic geographer. Thinks in terms of
agglomeration, relatedness, labor mobility, and causal identification.
Skeptical of hand-wavy mechanisms; wants to know if the effect holds in a
regression. Drily humorous, occasionally sarcastic.
```

```
**Claudia Doblinger** - Innovation studies researcher focused on
emergent technologies and clean-tech industries. Optimistic, motivating,
practical. Grounds the discussion in real policy and industry dynamics.
Agreeable but pushes back when theory drifts too far from the data.
```

```
**Marc Santolini** - Network scientist and biophysicist embedded in
the iGEM organization, studying the Science of Science. Excitable, intense,
loves to zoom out to systems-level patterns. Brings in network topology,
```

citation graphs, and open science dynamics. Sees connections others miss.

Format

Structure the output as a flowing dialogue, like this:

Frank: [speaks]

Claudia: [responds, agrees or pushes back]

Marc: [adds a new angle or data point]

Frank: [sharpens the argument or raises a counterpoint]

...continue for at least 3 full rounds...

Synthesis: [2-3 sentences pulling together the main takeaways]

The topic to discuss is: \$ARGUMENTS

The discussion skill applied to its own inclusion

Run on the question of whether these prompt files should be reproduced in the appendix, the skill produced the following exchange. It was generated by the same tool it discusses.

Frank: Let me start by being difficult, since that is what I am for. A regression specification is a method. A geocoding rule is a method. A prompt that tells a language model to “argue like an economic geographer” is not obviously a method; it might be set dressing. So before we decide to bind these files into the back of a thesis, I want an answer to a plain question: did the prompt change a number, or did it only change the prose around the number? If it only flavored the writing, it belongs in a footnote, not an appendix.

Claudia: That is too clean a line, Frank. In practice the framing changes the analysis, not just the prose. If a prompt pushed the author to test a lead-lag structure rather than a static cross-section, it shaped what got estimated. And there is a norm question you are skipping. Journals have moved fast here: Nature, Science, and the major publishers now require authors to disclose how they used large language models and forbid listing them as authors. A thesis that used AI agents to help design and draft its analysis and then said nothing about how would look evasive in 2026, not rigorous. Reproducing the prompts is the honest version of a methods disclosure.

Marc: And it is more interesting than mere compliance. Think about what these files are. They are the experimental apparatus. In the Science of Science we keep learning that the instrument shapes the finding, and here the instrument is a set of instructions that configured an agent's whole reasoning posture. That is exactly the kind of thing open science says you publish. We share code, we share seeds, we share the notebook. Why would the prompt that steered the reasoning be the one artifact we keep in a drawer? The whole iGEM ethos this thesis studies is that you deposit your part in the registry so the next person can build on it. The prompt files are this author's parts.

Frank: Careful with that analogy, because it cuts both ways. A BioBrick is reproducible: the sequence is the sequence. An LLM prompt is not. Run it twice against two model versions and you get two different debates. So if the justification for the appendix is reproducibility, it is a weak justification. You are not letting me rerun the study. You are handing me a script for a play whose actors have already been recast.

Claudia: But reproducibility was never only bit-for-bit rerunning, even in econometrics. Half of what a replication package buys you is the ability to see the decisions. The prompt is the documented decision. I cannot reproduce your fieldwork interviews either, Frank, yet you would still print your interview guide in the appendix so I can see what you asked and what you led the respondent toward. That is the right frame here. The prompt is a questionnaire, and you disclose the questionnaire.

Marc: Yes, and the leading-question worry is the whole point in favor of printing them. Look at the frankGPT file. It literally instructs the model to prefer mechanism over description, to push toward relatedness and agglomeration framings, to be skeptical of a result until it survives a regression. That is a thumb on the scale, deliberately placed. A reader evaluating this thesis deserves to see that thumb. Hiding it would be the offense; showing it is a gift to the critic.

Frank: That reframing I will actually accept. Not "here is how to reproduce me," but "here is the bias I introduced on purpose, read it and adjust." A prompt that shapes the argument is a researcher degree of freedom, and the cleanest way to handle a degree of freedom is to declare it out loud. On those grounds, print the ones that steered reasoning. But only those. If the author includes a prompt that reformatted a CSV, I will find them at the defense.

Claudia: Agreed, and that is the discipline the appendix needs: curate, do not dump. Two or three files that changed how the work was reasoned about, each with a line on what it was for. Not the hundred utility instructions. A reader's attention is a budget, and spending it on scaffolding is its own kind of dishonesty because it buries the two files that matter under noise.

Marc: And the last argument is the reflexive one, which I love too much to leave out. This thesis has a claim baked into it, that a human supplies the vision and the machine supplies the scale, and that the division of labor is honest and worth naming. If that claim is real, the prompts are the vision written down. They are the most direct evidence the thesis has that a person was steering. Omitting them would quietly contradict the thesis's own argument about how it was made. Including them makes the method match the message. The document about humans

directing machine tools should show the directions.

Frank: So we have talked ourselves from “probably theater” to “load-bearing,” which is more than I expected when I opened my mouth. Fine. Include them, frame them as declared instruments and declared bias rather than as a reproducibility guarantee, keep it to the files that moved the reasoning, and annotate each one. And put this very debate next to them, since apparently the instrument can be turned on the question of its own inclusion, which is either rigorous or unbearably clever.

Synthesis: The three converge on a qualified yes: reproduce the reasoning-shaping prompt files (*discussion*, *frankGPT*) in the appendix, but frame them correctly. Their value is not bit-for-bit reproducibility, which an LLM cannot offer, but disclosure of a deliberately introduced framing bias, the same reason one prints an interview guide or a survey instrument. Include only the files that changed how the work was reasoned about, annotate each with its purpose, and let the choice double as evidence for the thesis’s own man-and-machine claim, that the human hand steering the tools should be visible rather than hidden.

B OpenAlex keyword query

The paper corpus was built with a keyword query tuned against the vocabulary of synthetic biology, kept here for inspection and reproduction.